Sparse Innovation Regression with Compressed Sensing (SIR-CS)
Draft: Method and Synthetic Experiments

Abstract

We summarize a hybrid estimator that predicts a global component with machine learning and recovers a sparse innovation by compressed sensing under linear measurements. We then document a reproducible *Phase 0* synthetic baseline: ten independent seeds, measurement ratios up to $\rho = 0.6$, a refined ℓ_1 penalty grid for validation-based model selection, and standard diagnostics (curves versus ρ, gains over ml-only, parity plots, residual histograms, and example traces). A subsequent *Step 1* section compares FISTA versus SPGL1 on the same protocol; a *Step 1b* section adds `sklearn` Lasso and OMP baselines on identical (seed, ρ) draws with tables and figures from a frozen run. We also report a direct-regression benchmark that maps standardized $[u, b]$ to y (MLP, PCA+MLP, AE) against `hybrid_lfista_joint` on the same Phase 0 draws (frozen run `20260422_102236`): under identity Ψ and Gaussian M, the best direct maps achieve lower RMSE than joint LFISTA at most ρ, motivating regime-specific protocol extensions. A *Psi-axis* pilot (DCT Ψ with Gaussian M; run `20260422_175210`) tightens the joint-versus-MLP RMSE gap at high ρ but the best direct map still leads. A further *structured* direct benchmark with DCT Ψ and *subsampled* M (run `20260422_205913`) shows `hybrid_lfista_joint` achieving the best test MAE from $\rho = 0.30$ upward and the best RMSE/relative ℓ_2 at the largest ρ against the same $[u, b]$ baselines, illustrating a competitive regime for the SIR-CS joint map when the protocol matches background–innovation–partial measurement structure. Real-data evaluation is outlined as future work.

1 Sparse Innovation Regression with Compressed Sensing (SIR-CS)

1.1 Problem setup

Let $u \in \mathbb{R}^p$ denote the input variables and let $y \in \mathbb{R}^N$ be a high-dimensional target. At inference time, we observe only compressive measurements

$$b = My + \eta, \tag{1}$$

where $M \in \mathbb{R}^{m \times N}$, with $m \ll N$, is the measurement operator and η models measurement noise.

We decompose the target into a learned background component and an innovation term:

$$y = f_\theta(u) + r, \tag{2}$$

where $f_\theta : \mathbb{R}^p \to \mathbb{R}^N$ is a machine learning predictor and r is the residual. The main structural assumption is that the residual admits a sparse or compressible representation in a dictionary $\Psi \in \mathbb{R}^{N \times D}$:

$$r = \Psi\alpha^\star + \xi, \tag{3}$$

where $\alpha^\star \in \mathbb{R}^D$ is sparse (or compressible) and $\xi \in \mathbb{R}^N$ is a small non-sparse mismatch term.

Substituting this decomposition into the measurement model yields

$$z := b - Mf_\theta(u) = A\alpha^\star + \varepsilon, \qquad A := M\Psi, \qquad \varepsilon := M\xi + \eta. \tag{4}$$

Hence, inference reduces to recovering the innovation coefficients $\alpha^\star$ from the reduced measurements z, rather than reconstructing all N coordinates of y directly.

1.2 Inference: recovery of the innovation

The basic recovery stage is posed as a penalized sparse regression problem:

$$\hat{\alpha} \in \arg\min_{\alpha \in \mathbb{R}^D} \frac{1}{2}\|A\alpha - z\|_2^2 + \lambda\|\alpha\|_1. \tag{5}$$

The final estimate is then given by

$$\hat{y} = f_\theta(u) + \Psi\hat{\alpha}. \tag{6}$$

Structured variants. When support priors, spatial regularity, or domain constraints are available, we consider

$$\hat{\alpha} \in \arg\min_{\alpha \in \mathbb{R}^D} \frac{1}{2}\|A\alpha - z\|_2^2 + \lambda_1 \|W\alpha\|_1 + \lambda_2\, \Omega_{\mathrm{str}}(\alpha) + \lambda_3\, \mathcal{P}(\Psi\alpha), \quad (7)$$

where $W \in \mathbb{R}^{D\times D}$ is a diagonal weight matrix, Ω_{str} encodes structured sparsity (e.g., group, block, or tree penalties), and $\mathcal{P}$ encodes domain-specific regularity or physical constraints (e.g., total variation or physics-based penalties).

The basic estimator is recovered from (7) by taking $W = I$, $\lambda_2 = 0$, and $\lambda_3 = 0$.

1.3 Optional learned support prior

Let $g_\phi : \mathbb{R}^p \to [0,1]^D$ predict support-related scores

$$\pi(u) = (\pi_1(u), \ldots, \pi_D(u))^\top,$$

where $\pi_i(u)$ measures the confidence that the ith coefficient belongs to the support of $\alpha^\star$. A simple weighting scheme is

$$w_i = \begin{cases} \omega_{\mathrm{low}}, & \pi_i(u) \geq \tau, \\ 1, & \text{otherwise}, \end{cases} \qquad 0 < \omega_{\mathrm{low}} < 1, \quad \tau \in (0,1), \quad (8)$$

so that

$$\|W\alpha\|_1 = \sum_{i=1}^{D} w_i |\alpha_i|.$$

This choice penalizes less heavily those coefficients that are deemed more likely to belong to the innovation support.

1.4 Training objectives

Supervised training with sparse residual encouragement. When full targets y are available during training, one may estimate θ by minimizing

$$\mathcal{L}_{\mathrm{train}} = \|f_\theta(u) - y\|_2^2 + \beta\, \|\Psi^\top (y - f_\theta(u))\|_1 + \gamma\, \Omega_{\mathrm{str}}\big(\Psi^\top (y - f_\theta(u))\big), \quad (9)$$

when Ψ is orthonormal and an analysis representation is appropriate. More generally, $\Psi^\top r$ should be replaced by the analysis operator or latent representation associated with the chosen dictionary.

Task-aware training with simulated compression. Alternatively, one may simulate compressive observations

$$b = My + \eta$$

during training and minimize the post-recovery prediction error

$$\mathcal{L}_{\text{task}} = \|\hat{y}(u, b) - y\|_2^2, \tag{10}$$

possibly using unrolled proximal solvers or implicit differentiation when end-to-end differentiability is required.

1.5 Stability of the recovery stage (consequence of RIP)

Proposition 1 (Coefficient and reconstruction stability)**.** *Let $z = A\alpha^\star + \varepsilon$ with $A = M\Psi$ and $\|\varepsilon\|_2 \le \varepsilon_{\text{res}}$. Let $\widehat{\alpha}$ be any minimizer of $\|\alpha\|_1$ subject to $\|A\alpha - z\|_2 \le \varepsilon_{\text{res}}$ (equivalently, a BPDN solution at noise radius ε_{res}). Assume A satisfies Assumption 1 from Section 1.9 (equivalently, (24)). Let $\alpha_k^\star$ be the best k-sparse approximation to $\alpha^\star$ obtained by retaining the k largest magnitudes. Then there exist constants $C_0, C_1 > 0$ depending only on $(\delta_{3k}, \delta_{4k})$ such that*

$$\|\widehat{\alpha} - \alpha^\star\|_2 \le C_0 \, \frac{\|\alpha^\star - \alpha_k^\star\|_1}{\sqrt{k}} + C_1 \, \varepsilon_{\text{res}}. \tag{11}$$

If $\widehat{y} = f_\theta(u) + \Psi\widehat{\alpha}$ and $y = f^\star(u) + \Psi\alpha^\star + \xi$ with $f^\star$ the unknown background map, then

$$\|\widehat{y} - y\|_2 \le \|f_\theta(u) - f^\star(u)\|_2 + \|\Psi\|_{2\to 2} \, \|\widehat{\alpha} - \alpha^\star\|_2 + \|\xi\|_2. \tag{12}$$

Proof. The coefficient bound is Theorem 1 applied with $w = \varepsilon$ in (17); Appendix B gives the full proof under Assumption 1. For the reconstruction bound, write

$$\widehat{y} - y = \big(f_\theta(u) - f^\star(u)\big) + \Psi(\widehat{\alpha} - \alpha^\star) - \xi,$$

then apply the triangle inequality and the operator norm bound $\|\Psi v\|_2 \le \|\Psi\|_{2\to 2}\|v\|_2$. □

1.6 Algorithm

1.7 Method variants

V1 (SIR-CS): basic sparse recovery with ℓ_1 regularization on α.

Algorithm 1 SIR-CS inference

Require: Input u, measurements b, measurement operator M, predictor f_θ, dictionary Ψ

Require: Optional support-prior model g_ϕ

Require: Optional weights W, structured penalty Ω_{str}, and domain penalty $\mathcal{P}$

$\hat{y}_{\text{bg}} \leftarrow f_\theta(u)$

$z \leftarrow b - M\hat{y}_{\text{bg}}$

if a learned support prior is available **then**

    $\pi \leftarrow g_\phi(u)$

    Build the diagonal matrix $W = W(\pi)$

else

    $W \leftarrow I$

end if

Solve

$$\hat{\alpha} \in \arg\min_{\alpha \in \mathbb{R}^D} \frac{1}{2}\|M\Psi\alpha - z\|_2^2 + \lambda_1\|W\alpha\|_1 + \lambda_2\Omega_{\text{str}}(\alpha) + \lambda_3\mathcal{P}(\Psi\alpha)$$

$\hat{y} \leftarrow \hat{y}_{\text{bg}} + \Psi\hat{\alpha}$

Ensure: $\hat{y}$

V2 (WSIR-CS): weighted sparse recovery using a diagonal matrix W induced by support priors.

V3 (TSIR-CS): temporal regularization on α_t, for example through a term such as $\|\alpha_t - \hat{\alpha}_{t-1}\|_2^2$.

V4 (Joint-SIR-CS): joint recovery with a shared component α_c and channel-specific innovations α_j across multiple outputs or sensing channels.

1.8 Task-aware unrolled proximal refinement (LFISTA)

The classical recovery step in Section 1 uses a generic first-order sparse solver (e.g., FISTA) with fixed iteration schedule and hand-tuned λ. An alternative is to *unroll* a fixed number K of proximal-gradient steps on the innovation subproblem $\min_\alpha \frac{1}{2}\|A\alpha - z\|_2^2 + \lambda\|\alpha\|_1$ with $A = M\Psi$ and $z = b - Mf_\theta(u)$, and to treat step sizes and soft-threshold levels as *learnable* per layer. We denote this block LFISTA (learned FISTA): it maps z to $\hat{\alpha}^{(K)}$ and yields $\hat{y} = f_\theta(u) + \Psi\hat{\alpha}^{(K)}$. The background f_θ is a multi-layer perceptron in PyTorch (not identical to the sklearn MLP used for `ml_only` in the pipeline).

Training stages (implementation). (i) Background: train f_θ by minimizing $\|f_\theta(u) - y\|_2^2$ on the training set. **(ii) Frozen LFISTA:** fix f_θ, train only the unrolled proximal parameters to minimize $\|\hat{y} - y\|_2^2$. **(iii) Joint:** fine-tune f_θ and the LFISTA block together with a smaller learning rate. This mirrors task-aware objectives in Section 1 but replaces the inner convex solver by a differentiable unrolled map.

1.9 Theoretical perspective

We work with a synthesis dictionary $\Psi \in \mathbb{R}^{N\times N}$ that is orthonormal, i.e. $\Psi^\top\Psi = I$. This is the standard setting in much of the compressed sensing literature and covers common choices such as the identity basis, orthonormal DCT, or unitary DFT models under the usual normalizations. The specialization $\Psi = I$ is the familiar coordinate representation. Elsewhere in this draft the implementation exposition often writes $\Psi \in \mathbb{R}^{N\times D}$; here we take $D = N$ so that $\alpha^\star \in \mathbb{R}^N$.

1.9.1 Structured model and measurements

Let

$$y = f^\star(u) + \Psi\alpha^\star + \xi, \tag{13}$$

where:

- $y \in \mathbb{R}^N$ is the target signal;
- u denotes the contextual covariates;
- $f^\star(u)$ is the unknown background (global) component;
- $\Psi \in \mathbb{R}^{N \times N}$ is orthonormal;
- $\alpha^\star \in \mathbb{R}^N$ is a sparse or compressible innovation coefficient vector;
- $\xi \in \mathbb{R}^N$ is output noise or model mismatch.

Compressed observations are

$$b = My + \eta, \tag{14}$$

where $M \in \mathbb{R}^{m \times N}$ is the measurement operator and $\eta \in \mathbb{R}^m$ is measurement noise.

The hybrid predictor estimates a background $f_\theta(u)$ and forms the measurement-space residual

$$z := b - Mf_\theta(u). \tag{15}$$

1.9.2 Residual reduction identity

Define the background approximation error

$$e_{\mathrm{bg}}(u) := f^\star(u) - f_\theta(u). \tag{16}$$

Proposition 2 (Residual reduction under an orthonormal dictionary)**.** *Under* (13)–(15)*, the measurement-space residual satisfies*

$$z = A\alpha^\star + w, \qquad A := M\Psi, \qquad w := Me_{\mathrm{bg}}(u) + M\xi + \eta. \tag{17}$$

Proof. By (14) and (13),

$$b = M\big(f^\star(u) + \Psi\alpha^\star + \xi\big) + \eta.$$

Subtracting $Mf_\theta(u)$ and using (15),

$$\begin{aligned} z &= b - Mf_\theta(u) \\ &= M\big(f^\star(u) + \Psi\alpha^\star + \xi\big) + \eta - Mf_\theta(u) \\ &= M\Psi\alpha^\star + M\big(f^\star(u) - f_\theta(u)\big) + M\xi + \eta. \end{aligned}$$

Using (16) gives (17). □

Interpretation. Equation (17) is the key algebraic reduction of the hybrid construction: the sparse stage no longer attempts to recover the whole signal y, but only the innovation coefficients $\alpha^\star$ through a noisy inverse problem with sensing matrix $A = M\Psi$. If $\|e_{\mathrm{bg}}(u)\|_2$ is small and M does not amplify it too strongly, then $\|w\|_2$ is small (other terms fixed), which improves the conditioning of the residual problem relative to naive reconstruction of y.

1.9.3 Deterministic decomposition of the final reconstruction error

Let $\widehat{\alpha}$ be any estimate of $\alpha^\star$, and define

$$\widehat{y} := f_\theta(u) + \Psi\widehat{\alpha}. \tag{18}$$

Proposition 3 (Deterministic decomposition of the final error)**.** *Under* (13) *and* (18)*,*

$$\widehat{y} - y = -e_{\mathrm{bg}}(u) + \Psi(\widehat{\alpha} - \alpha^\star) - \xi. \tag{19}$$

Consequently,

$$\|\widehat{y} - y\|_2 \le \|e_{\mathrm{bg}}(u)\|_2 + \|\Psi(\widehat{\alpha} - \alpha^\star)\|_2 + \|\xi\|_2. \tag{20}$$

Since Ψ *is orthonormal,*

$$\|\widehat{y} - y\|_2 \le \|e_{\mathrm{bg}}(u)\|_2 + \|\widehat{\alpha} - \alpha^\star\|_2 + \|\xi\|_2. \tag{21}$$

Proof. The identity (19) follows by direct substitution of (13) and (18), together with $e_{\mathrm{bg}}(u) = f^\star(u) - f_\theta(u)$. Inequality (20) is the triangle inequality, and (21) uses $\|\Psi v\|_2 = \|v\|_2$ for all v when Ψ is orthonormal. □

Remark on the two roles of the background error. The background error appears in two distinct ways:

- as $\|e_{\mathrm{bg}}(u)\|_2$ in the signal-domain reconstruction error;
- through $\|Me_{\mathrm{bg}}(u)\|_2$ inside the effective perturbation of the residual inverse problem (embedded in w in (17)).

These are not redundant: the first measures mismatch in the target space, whereas the second measures how that mismatch is seen through the measurement operator.

1.9.4 Stable sparse recovery for the hybrid residual problem (classical estimator)

We analyze the sparse stage with a standard convex estimator. Consider constrained BPDN,

$$\widehat{\alpha} \in \arg\min_{\alpha \in \mathbb{R}^N} \|\alpha\|_1 \quad \text{subject to} \quad \|A\alpha - z\|_2 \le \varepsilon, \tag{22}$$

with $A = M\Psi$. Assume the effective noise bound

$$\|w\|_2 \le \varepsilon, \tag{23}$$

for w from (17).

Assumption 1 (Restricted isometry)**.** *The matrix $A = M\Psi$ satisfies the restricted isometry property up to order $4k$ with constants $(\delta_s)_{s \le 4k}$ such that*

$$\delta_{3k} + 3\delta_{4k} < 2, \tag{24}$$

as in Candès, Romberg, and Tao (2005, Theorem 2). In particular, (24) *holds whenever $\delta_{4k} \le 1/5$.*

Comment on plausibility. In the synthetic protocol of this draft, M is Gaussian and the baseline innovation basis is orthonormal (identity or DCT). In such regimes, (24) is a standard sufficient hypothesis with high probability for m large enough relative to k and N; we do not verify RIP numerically for each realization.

Under Assumption 1, one obtains the following stability guarantee for the *classical* hybrid residual stage (22).

Theorem 1 (Stable sparse recovery for the hybrid residual problem)**.** *Let $\alpha^\star \in \mathbb{R}^N$ be arbitrary and let $\alpha_k^\star$ denote its best k-sparse approximation obtained by retaining the k largest magnitudes. Suppose Assumption 1 holds and let $\widehat{\alpha}$ solve* (22) *with ε satisfying* (23)*. Then there exist constants $C_0, C_1 > 0$, depending only on $(\delta_{3k}, \delta_{4k})$, such that*

$$\|\widehat{\alpha} - \alpha^\star\|_2 \le C_0 \frac{\|\alpha^\star - \alpha_k^\star\|_1}{\sqrt{k}} + C_1 \varepsilon. \tag{25}$$

Using (17)*, one may take*

$$\varepsilon = \|M e_{\mathrm{bg}}(u) + M\xi + \eta\|_2. \tag{26}$$

Therefore,

$$\|\widehat{\alpha} - \alpha^\star\|_2 \le C_0 \frac{\|\alpha^\star - \alpha_k^\star\|_1}{\sqrt{k}} + C_1 \|M e_{\mathrm{bg}}(u) + M\xi + \eta\|_2. \tag{27}$$

Proof. By Proposition 2,

$$z = A\alpha^\star + w, \qquad A = M\Psi, \qquad w = Me_{\mathrm{bg}}(u) + M\xi + \eta.$$

Under (23), the vector $\alpha^\star$ is feasible for (22) because $\|A\alpha^\star - z\|_2 = \|w\|_2 \le \varepsilon$. The remainder of the argument is the standard compressible-signal stability analysis for BPDN under RIP: under (24), classical compressed sensing theory yields constants $C_0, C_1 > 0$ depending only on $(\delta_{3k}, \delta_{4k})$ such that (25) holds for every minimizer $\widehat{\alpha}$ of (22). Appendix B records a self-contained proof specialized to the hybrid residual model (Candès–Romberg–Tao, Section 2.2). Identifying ε with $\|w\|_2$ yields (26)–(27). □

1.9.5 Final reconstruction bound for the hybrid estimator

Corollary 1 (Hybrid reconstruction bound under an orthonormal dictionary)**.** *Under the assumptions of Theorem 1, the hybrid reconstruction* $\widehat{y} = f_\theta(u) + \Psi\widehat{\alpha}$ *satisfies*

$$\|\widehat{y} - y\|_2 \le \|e_{\mathrm{bg}}(u)\|_2 + C_0 \frac{\|\alpha^\star - \alpha_k^\star\|_1}{\sqrt{k}} + C_1 \|Me_{\mathrm{bg}}(u) + M\xi + \eta\|_2 + \|\xi\|_2. \quad (28)$$

Proof. By (21),

$$\|\widehat{y} - y\|_2 \le \|e_{\mathrm{bg}}(u)\|_2 + \|\widehat{\alpha} - \alpha^\star\|_2 + \|\xi\|_2.$$

Apply (27) to $\|\widehat{\alpha} - \alpha^\star\|_2$ to obtain (28). □

Interpretation. The bound (28) shows that the total error is controlled by four contributions:

1. the background prediction error $\|e_{\mathrm{bg}}(u)\|_2$;
2. the compressibility defect $\|\alpha^\star - \alpha_k^\star\|_1/\sqrt{k}$;
3. the effective perturbation in the residual inverse problem, $\|Me_{\mathrm{bg}}(u) + M\xi + \eta\|_2$;
4. the direct output mismatch $\|\xi\|_2$.

Thus the hybrid strategy is theoretically favorable when the contextual predictor explains a substantial fraction of the signal energy and the remaining innovation is sparse or compressible in the chosen orthonormal basis.

1.9.6 Particular case: identity basis

Corollary 2 (Identity basis as a particular case)**.** *Assume* $\Psi = I$*. Then* (13) *becomes*

$$y = f^\star(u) + \alpha^\star + \xi, \tag{29}$$

the effective sensing matrix is $A = M$*, and* (17) *reads*

$$z = M\alpha^\star + Me_{\mathrm{bg}}(u) + M\xi + \eta. \tag{30}$$

Moreover, the specialization of (28) *reads explicitly as*

$$\|\widehat{y} - y\|_2 \le \|e_{\mathrm{bg}}(u)\|_2 + C_0 \frac{\|\alpha^\star - \alpha_k^\star\|_1}{\sqrt{k}} + C_1 \|Me_{\mathrm{bg}}(u) + M\xi + \eta\|_2 + \|\xi\|_2. \tag{31}$$

Proof. Set $\Psi = I$ in (13), (17), and (28). Since I is orthonormal, all intermediate results apply directly. □

Examples covered by the orthonormal case. The orthonormal template directly covers $\Psi = I$ (canonical coefficients), orthonormal DCT bases, and unitary DFT synthesis models under standard normalizations. Therefore the same theoretical picture applies whenever the innovation is sparse or compressible in one of these orthonormal transform domains.

1.9.7 A comparison with pure compressed sensing

If one ignores the contextual term and treats y itself as sparse in some basis, the coefficient vector must simultaneously explain background and innovation. The hybrid formulation isolates the innovation in (17), which is better matched to structured signals whenever the background is not sparse in the same basis.

1.9.8 Structural statements for the joint LFISTA model

The joint LFISTA predictor takes the form

$$\widehat{y}_{\mathrm{joint}}(u, b) = f_\theta(u) + \Psi\, \Phi_\phi(M\Psi,\ b - M f_\theta(u)), \tag{32}$$

where Φ_ϕ is a K-layer unrolled proximal map.

For $x \in \mathbb{R}^N$ and $\tau \ge 0$, let $\mathcal{S}_\tau : \mathbb{R}^N \to \mathbb{R}^N$ denote coordinatewise soft-thresholding, $(\mathcal{S}_\tau(x))_i = \mathrm{sign}(x_i)\max(|x_i| - \tau, 0)$. A typical layer is

$$\alpha^{t+1} = \mathcal{S}_{\tau_t}\Big(v^t - \eta_t A^\top (Av^t - z)\Big), \qquad A = M\Psi, \tag{33}$$

possibly with FISTA-style momentum in v^t.

For this learned model, the strongest fully rigorous statements are *structural*, not uniform performance guarantees relative to a tuned convex solver.

Remark 1 (Joint versus frozen training)**.** *Let $\mathcal{L}(\theta, \phi)$ denote the training objective of the joint LFISTA model, where θ indexes the background network and ϕ indexes the unrolled sparse block. Let Θ and Φ be the admissible parameter sets. The frozen problem minimizes $\mathcal{L}$ over $\{\theta_0\} \times \Phi$, whereas the joint problem minimizes over $\Theta \times \Phi$. Since $\{\theta_0\} \times \Phi \subseteq \Theta \times \Phi$, taking infima yields*

$$\inf_{\theta,\phi} \mathcal{L}(\theta, \phi) \leq \inf_{\phi} \mathcal{L}(\theta_0, \phi). \tag{34}$$

This is a statement about feasible model class *only; it does not assert that gradient-based training reaches a better minimum in practice.*

1.9.9 Finite-depth LFISTA: a triangle decomposition

Let $\alpha^\infty(z)$ denote an ideal iterative limit associated with the residual inverse problem, and let $\alpha_\phi^K(z)$ denote the output of a K-layer unrolled map. By the triangle inequality,

$$\|\alpha_\phi^K(z) - \alpha^\star\|_2 \leq \|\alpha_\phi^K(z) - \alpha^\infty(z)\|_2 + \|\alpha^\infty(z) - \alpha^\star\|_2, \tag{35}$$

which separates a finite-depth approximation term from the inverse-problem error of the ideal limit.

Consequence for the LFISTA reconstruction. Let $\widehat{y}_{\mathrm{LF}} := f_\theta(u) + \Psi\alpha_\phi^K(z)$ with z from (15). If Ψ is orthonormal, combining (21) with (35) gives

$$\|\widehat{y}_{\mathrm{LF}} - y\|_2 \leq \|e_{\mathrm{bg}}(u)\|_2 + \|\alpha_\phi^K(z) - \alpha^\infty(z)\|_2 + \|\alpha^\infty(z) - \alpha^\star\|_2 + \|\xi\|_2. \tag{36}$$

This is a purely deterministic bookkeeping inequality; it is not a statistical performance guarantee for the learned map Φ_ϕ.

Scope. Inequality (36) separates (i) background error, (ii) finite-depth truncation, (iii) error of the ideal limit, and (iv) output noise. It does *not* imply global optimality of training, uniform superiority over convex solvers, preservation of exact RIP-type guarantees after learning, or optimal generalization.

1.9.10 Summary of the theoretical picture

The strongest rigorous guarantees in this subsection apply to the **classical hybrid** residual formulation with a standard sparse estimator under Assumption 1: Proposition 2, Proposition 3, and Theorem 1 together control the reconstruction error in terms of background mismatch in *both* the signal domain and the measurement domain (via w), compressibility of $\alpha^\star$, and noise.

For **joint LFISTA**, the rigorous statements are structural and deterministic: Remark 1, the triangle decomposition (35), and the bookkeeping inequality (36). They justify residual reduction, the larger feasible class of joint training relative to frozen training, and a decomposition of the learned error into truncation and inverse-problem terms.

What cannot be claimed without further assumptions. One generally cannot assert that the learned joint LFISTA model:

- reaches a global optimum during training;
- uniformly outperforms classical convex sparse solvers;
- preserves exact sparse recovery guarantees after learning;
- generalizes optimally outside the training distribution.

Empirical gains therefore require hypotheses beyond this deterministic skeleton.

1.10 Empirical comparison: LFISTA versus classical hybrid on synthetic data

We report a controlled benchmark that keeps the Phase 0 data-generating process and the same ten seeds $\{7, 13, 23, 29, 31, 37, 41, 43, 47, 53\}$ and measurement ratios $\rho \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$ as in the main synthetic section. The pipeline enables *simultaneously*: (a) classical SIR-CS with FISTA on the residual, exposed as `hybrid_fista`, with SPGL1 and CS-only variants for context (same dual-solver protocol as Step 1); (b) the PyTorch branch `ml_only_torch`, `hybrid_lfista_frozen`, `hybrid_lfista_joint`. Thus `hybrid_fista` and the LFISTA variants share the same M, Ψ, splits, and noise model per job; only the estimator class differs. Figures below are taken from the folder `figures/lfista_vs_classical/` (generated by `sir_cs_pipeline_optimized.py` with profile `lfista_vs_classical`).

Table 1: Test RMSE (mean ± std across ten seeds) for the classical FISTA hybrid versus LFISTA and PyTorch baselines, versus $\rho = m/N$. Best RMSE in each row among `hybrid_fista`, `hybrid_lfista_frozen`, and `hybrid_lfista_joint` in bold.

ρ	`hybrid_fista`	`hybrid_lfista_frozen`	`hybrid_lfista_joint`
0.20	0.336 ± 0.011	$\mathbf{0.317} \pm 0.010$	0.317 ± 0.010
0.30	0.319 ± 0.011	$\mathbf{0.299} \pm 0.007$	$\mathbf{0.299} \pm 0.007$
0.40	0.296 ± 0.011	0.273 ± 0.008	$\mathbf{0.273} \pm 0.008$
0.50	0.272 ± 0.010	0.254 ± 0.008	$\mathbf{0.253} \pm 0.009$
0.60	0.245 ± 0.010	0.230 ± 0.009	$\mathbf{0.228} \pm 0.009$

Reference baselines (same run): sklearn `ml_only` RMSE ≈ 0.369 (flat in ρ); `ml_only_torch` RMSE ranges from 0.362 (low ρ) to 0.374 (high ρ).

Reading Table 1. Both LFISTA variants achieve lower RMSE than `hybrid_fista` at every ρ, with relative reductions of roughly 5–8% versus `hybrid_fista` depending on ρ. Frozen and joint estimates are nearly tied at low ρ; joint becomes slightly better at $\rho \geq 0.5$, consistent with end-to-end adaptation of f_θ when measurements are more informative. Uncertainty (std across seeds) for LFISTA is comparable to or smaller than for `hybrid_fista`, so the gap is not explained by higher seed variance. The PyTorch-only baseline `ml_only_torch` remains above `hybrid_fista`, confirming that residual CS (classical or learned) is critical in this regime.

1.10.1 Error curves and gains

Figure 1 shows RMSE, MAE, and relative ℓ_2 error versus ρ for all methods in the run (including SPGL1 hybrids and CS-only baselines). The ordering of the central estimators matches Table 1: LFISTA curves lie below `hybrid_fista`; `ml_only_torch` tracks between sklearn `ml_only` and the hybrid methods.

Figure 2 (left) reports support F1 where defined; Figure 2 (right) shows grouped RMSE by ρ. Support F1 stays in a narrow band, so discrimination is driven primarily by reconstruction error rather than support overlap.

Figure 3 summarizes gains over sklearn `ml_only` (RMSE and MAE); all hybrid estimators remain positive, with LFISTA and `hybrid_fista` largest. Figure 4 shows the analogous gains over `ml_only_torch`: LFISTA variants and `hybrid_fista` remain positive, quantifying improvement over the Py-

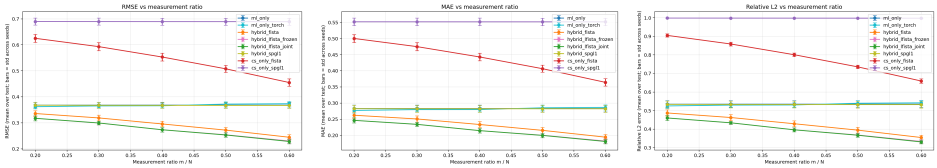


Figure 1: Test error versus ρ for the `lfista_vs_classical` run: classical and learned hybrids, SPGL1 variants, CS-only, and ML baselines. Bars: std across ten seeds.

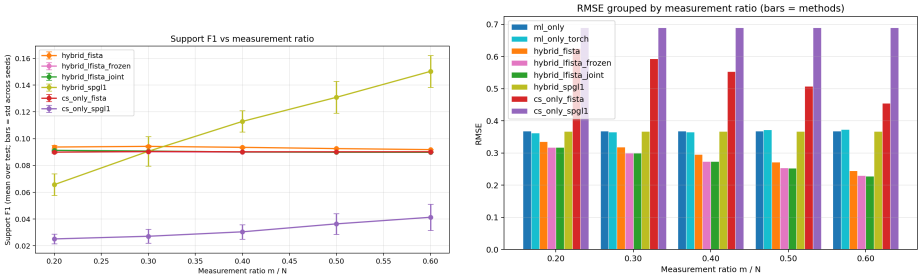


Figure 2: Left: support F1 versus ρ. Right: grouped RMSE by ρ (`lfista_vs_classical`).

Torch background alone.

1.10.2 Parity plots, residuals, and example traces

Figure 5 pools predictions across test samples, seeds, and ρ. Figure 6 shows residual histograms; Figure 7 shows example output-index traces. The learned hybrids track the identity line tightly; CS-only and pure ML baselines show the same qualitative gaps as in Phase 0.

Discussion. The synthetic evidence supports two statements aligned with the roadmap: (i) LFISTA improves over a PyTorch background without CS (Figure 4); (ii) under this protocol, unrolled proximal training can *lower* test RMSE relative to a strong classical FISTA hybrid on the same residual (Table 1). This does not imply superiority on all datasets or measurement models: the comparison omits, for example, identical-z evaluation of classical FISTA against the PyTorch forward (a finer diagnostic left for future work), and computational cost of training is substantially higher than for classical solvers.

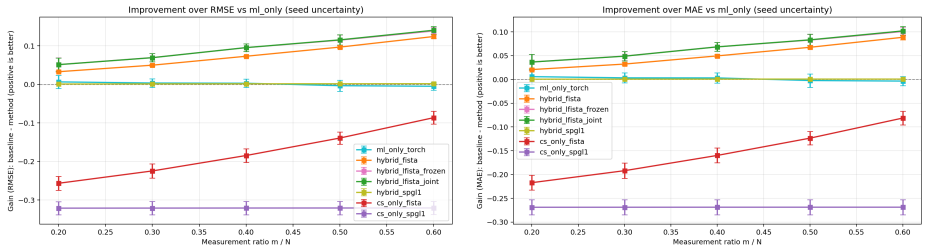


Figure 3: Gain over `ml_only` (positive is better): `lfista_vs_classical` run.

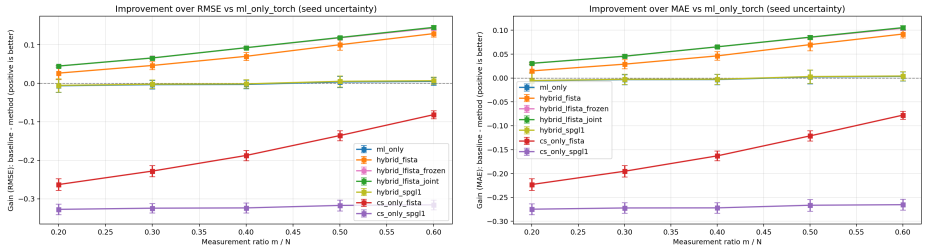


Figure 4: Gain over `ml_only_torch` (positive is better). Classical and LFISTA hybrids improve over the PyTorch MLP baseline.

Reproducibility. The run is produced by `python sir_cs_pipeline_optimized.py --profile lfista_vs_classical`; artefacts include `summary.csv`, `summary_focus_ml_hybrid_fista_l` and per-run logs under `outputs/lfista_vs_classical/runs/`. PNG exports for this draft are mirrored under `paper/figures/lfista_vs_classical/`.

2 Experiments: synthetic validation (Phase 0 baseline)

This section reports a *Phase 0* controlled synthetic study: a frozen protocol aligned with the project's ablation roadmap. The goal is to verify that the SIR-CS pipeline behaves consistently with Section 1 in a favorable regime (learnable global component, sparse innovation in the recovery basis, Gaussian compressive sensing). Uncertainty is quantified across *independent* random seeds (not pseudo-replications over test samples). A complementary evaluation on *real data* remains future work (Section 2.8).

2.1 Protocol

Phase 0 reference configuration. We fix the data-generating process and training protocol, then aggregate statistics across ten seeds $\{7, 13, 23, 29, 31, 37, 41, 43, 47, 53\}$. Measurement ratios are $\rho \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$ with $N = 128$ and $m = \lfloor \rho N \rfloor$. All other hyperparameters match the implementation defaults for this profile: $p = 12$, $k = 6$, $\Psi = I$, `residual_mode`=`support_from_u`, Gaussian M, `measurement_noise_std`=0.02, `output_noise_std`=0.01.

Data generation. Inputs $u \in \mathbb{R}^p$ are drawn i.i.d. standard Gaussian. Targets satisfy $y = y_{\text{bg}} + \Psi\alpha^\star + \xi_{\text{out}}$ where y_{bg} is a smooth nonlinear random-feature map of u, $\alpha^\star$ is k-sparse in canonical coordinates, $\Psi = I$, and ξ_{out} is small output noise.

Train and test splits. We use $n_{\text{train}} = 1200$, $n_{\text{val}} = 300$, and $n_{\text{test}} = 300$ samples per seed.

Measurement model. Observations are $b = My + \eta$ with additive Gaussian measurement noise at fixed standard deviation, shared consistently between validation-based λ selection and test evaluation in the code.

Methods. **ml_only:** multi-output MLP predicting y from u without using M at inference. **hybrid:** SIR-CS with FISTA/LASSO on the innovation after subtracting $Mf_\theta(u)$ from b. **weighted_hybrid:** same as hybrid with diagonal weights in the ℓ_1 term derived from an auxiliary predictor of coefficient magnitudes. **cs_only:** sparse reconstruction of y in Ψ from b without the learned background (misspecified reference).

Hyperparameters and model selection. The ℓ_1 penalty is selected on the validation set over the grid $\{10^{-4}, 3\cdot10^{-4}, 10^{-3}, 3\cdot10^{-3}, 10^{-2}, 3\cdot10^{-2}\}$. Lipschitz constants for FISTA are precomputed once per (M, Ψ). Reported numbers aggregate *first* over test samples within each (seed, ρ), then *across seeds* as mean $\pm$ standard deviation.

2.2 Main quantitative results

Table 2 summarizes test RMSE and MAE. The ml-only baseline is *flat* in ρ because its prediction does not use the realized compressive measurements at test time.

Table 2: Test RMSE and MAE (mean ± std across ten seeds) versus $\rho = m/N$. Best RMSE in each row in bold (excluding `cs_only`).

ρ	`ml_only`	`hybrid`	`weighted_hybrid`	`cs_only`
RMSE				
0.20	0.369 ± 0.011	$\mathbf{0.336} \pm 0.011$	0.340 ± 0.011	0.625 ± 0.015
0.30	0.369 ± 0.011	$\mathbf{0.319} \pm 0.011$	0.325 ± 0.011	0.593 ± 0.016
0.40	0.369 ± 0.011	$\mathbf{0.296} \pm 0.011$	0.305 ± 0.011	0.553 ± 0.015
0.50	0.369 ± 0.011	$\mathbf{0.272} \pm 0.011$	0.283 ± 0.010	0.508 ± 0.013
0.60	0.369 ± 0.011	$\mathbf{0.245} \pm 0.010$	0.258 ± 0.010	0.455 ± 0.015
MAE				
0.20	0.283 ± 0.011	$\mathbf{0.262} \pm 0.010$	0.266 ± 0.010	0.500 ± 0.012
0.30	0.283 ± 0.011	$\mathbf{0.251} \pm 0.010$	0.256 ± 0.010	0.475 ± 0.013
0.40	0.283 ± 0.011	$\mathbf{0.234} \pm 0.009$	0.241 ± 0.009	0.443 ± 0.012
0.50	0.283 ± 0.011	$\mathbf{0.215} \pm 0.009$	0.225 ± 0.009	0.407 ± 0.011
0.60	0.283 ± 0.011	$\mathbf{0.194} \pm 0.008$	0.205 ± 0.008	0.364 ± 0.012

Reading Table 2. `hybrid` achieves the lowest RMSE and MAE at every ρ. The margin over ml-only grows monotonically with ρ in this table: RMSE drops from 0.369 to 0.245 at $\rho = 0.60$ (absolute reduction 0.124 relative to ml-only at the same ρ). `cs_only` remains far worse because it omits the global component $f_\theta(u)$. `weighted_hybrid` improves consistently over ml-only but remains above `hybrid`; the RMSE gap `weighted_hybrid`−`hybrid` widens slightly as ρ increases, which motivates dedicated prior diagnostics (support correlation, oracle weights) rather than treating the weighted variant as the primary estimator in this regime.

2.3 Error versus measurement ratio

Figure 8 shows RMSE, MAE, and relative ℓ_2 error versus ρ. Error bars show the standard deviation across seeds.

Discussion. The monotone improvement of `hybrid` with ρ matches the inverse-problem interpretation in Section 1: more linear measurements improve conditioning of the effective operator $M\Psi$ at fixed noise. The flat ml-only curve is expected: the baseline ignores M at prediction time. Relative ℓ_2 tracks RMSE/MAE: `cs_only` saturates at high error, while hybrid methods follow the same ordering as in Table 2.

2.4 Structured recovery metrics and grouped RMSE

Figure 9 (left) reports support F1 for methods that yield sparse coefficients comparable to ground truth. `ml_only` is omitted. Figure 9 (right) shows grouped RMSE by ρ.

Discussion. Support F1 stays in a narrow band for all sparse methods, with hybrid methods slightly above `cs_only`; this metric should be read jointly with reconstruction error, because high F1 alone does not rescue `cs_only` under misspecification. The bar panel mirrors Table 2: decreasing hybrid error with ρ, flat ml-only, and elevated `cs_only`.

2.5 Gains over the ml-only baseline

Figure 10 shows per-seed gains $\mathrm{RMSE}_{\mathrm{ml}} - \mathrm{RMSE}_{\mathrm{method}}$ (and MAE analog), aggregated across seeds. Positive values mean improvement over ml-only.

Discussion. `hybrid` and `weighted_hybrid` yield uniformly positive gains in this Phase 0 window, with larger gains at higher ρ. `hybrid` dominates `weighted_hybrid` in both panels, consistent with Table 2. `cs_only` remains strongly negative, quantifying the mismatch when the dense background is ignored.

2.6 Parity plots, residuals, and example traces

Figure 11 pools test coordinates across seeds and ρ. Figure 12 shows histograms of $\hat{y} - y$. Figure 13 shows example traces along the output index.

Discussion. Parity plots provide a global calibration check: `hybrid` tracks the identity line most closely, while `cs_only` exhibits the classic artifact of a misspecified sparse model under dense targets. Residual histograms and traces reinforce the quantitative ordering in Table 2.

2.7 Limitations

These results establish internal consistency and a favorable synthetic regime. They do *not* guarantee performance under arbitrary real-world conditions (non-Gaussian M, basis mismatch, ill-conditioned subsampling operators, heavy-tailed noise, or poorly calibrated λ grids). In validation logs for

Table 3: Frozen runs for this section: residual basis Ψ, measurement operator kind for M, and figure subfolder under `paper/figures/`.

`run_id`	Ψ	M	subfolder of `paper/figures/`
`20260422_102236`	identity	Gaussian	`direct_ub_benchmark/`
`20260422_175210`	DCT	Gaussian	`direct_ub_psi_dct/`
`20260422_205913`	DCT	subsample	`direct_ub_dct_subsample/`

Phase 0, the selected λ often lies at the *smallest* grid value (10^{-4}), suggesting that future sweeps could extend the grid toward smaller penalties if computational budget allows.

2.8 Outlook: evaluation on real data

The next experimental milestone is to instantiate the same comparison template on a real dataset with physically motivated M and Ψ (for example, a field subsampled at sensor locations). The Phase 0 section then serves as a reproducible sanity check and ablation reference, while the real-data study targets external validity and domain baselines.

3 Experiments: direct $[u, b] \to y$ baselines versus joint LFISTA

This section complements Phase 0 (Section 2) with an *end-to-end* comparison on the **same** synthetic generator and (seed, ρ) protocol: several regressors predict y directly from standardized $[u, b]$ with $b = My + \eta$, versus the PyTorch joint LFISTA map (`hybrid_lfista_joint`) trained in the LFISTA branch of the pipeline. We first document the Phase 0-style regime ($\Psi = I$, Gaussian M), then two controlled ablations (Ψ axis, then M axis at DCT Ψ), ending with a structured DCT+subsample protocol (Section 3.5) where the joint estimator becomes strongly competitive. The initial goal is to show when a strong direct nonlinear map can outperform the structured path; the later subsections show when the opposite holds under a more SIR-CS-aligned measurement model. Table 3 lists the three frozen run identifiers used below (same script and (seed, ρ) protocol unless noted).

Table 4: Test RMSE (mean ± std across ten seeds) for direct $[u, b] \to y$ baselines versus `hybrid_lfista_joint`. Frozen run `20260422_102236`.

ρ	`ml_only`	`mlp_concat_ub`	`pca_regression_ub`	`ae_regression_ub`	`hybrid_lfista_joint`
0.20	0.369 ± 0.011	0.283 ± 0.007	$\mathbf{0.282 \pm 0.006}$	0.290 ± 0.007	0.317 ± 0.008
0.30	0.369 ± 0.011	0.255 ± 0.005	$\mathbf{0.255 \pm 0.005}$	0.266 ± 0.005	0.299 ± 0.008
0.40	0.369 ± 0.011	$\mathbf{0.228 \pm 0.005}$	0.232 ± 0.004	0.250 ± 0.005	0.276 ± 0.007
0.50	0.369 ± 0.011	$\mathbf{0.208 \pm 0.003}$	0.212 ± 0.004	0.239 ± 0.005	0.253 ± 0.006
0.60	0.369 ± 0.011	$\mathbf{0.190 \pm 0.004}$	0.196 ± 0.004	0.231 ± 0.004	0.228 ± 0.006

3.1 Protocol

We run `sir_cs_benchmark_direct_ub.py` with profile `direct_ub_lfista_joint_only`: seeds $\{7, 13, \ldots, 53\}$, $\rho \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$, identical `make_dataset` splits and measurement draws as the hybrid evaluation for each job. Features are $[u, b]$ with independent Gaussian noise on b at train/val/test; `StandardScaler` is fit on train $[u, b]$ only.

Methods. `ml_only`: sklearn MLP on u only (reference; flat in ρ). `mlp_concat_ub`: MLP on scaled $[u, b]$ (same hidden layers and regularization defaults as the pipeline MLP baseline). `pca_regression_ub`: PCA on Y_{train} only; latent dimension selected by validation; MLP maps $[u, b]$ to PCA scores then inverse-transforms to y. `ae_regression_ub`: PyTorch autoencoder on standardized Y_{train}; MLP maps $[u, b]$ to latent then decodes to y. `hybrid_lfista_joint`: joint LFISTA training (`run_lfista_branch`); this run omits `hybrid_fista`, `ml_only_torch`, and `hybrid_lfista_frozen`.

3.2 RMSE summary

Table 4 reports test RMSE (mean ± std across ten seeds). The lowest RMSE in each row is in bold.

Reading Table 4. Under the Phase 0 generator with identity Ψ and Gaussian M, the direct regressors `mlp_concat_ub` and `pca_regression_ub` achieve the best RMSE at every ρ in this table; `hybrid_lfista_joint` improves monotonically with ρ and nearly matches `ae_regression_ub` at $\rho = 0.60$, but does not beat the best direct map in this window. `ml_only` remains flat in ρ by construction (no b at test time). These numbers support treating the current synthetic instance primarily as a regime where global

Table 5: Test RMSE (mean over ten seeds) for `hybrid_lfista_joint` and `mlp_concat_ub` when $\Psi = I$ (run `20260422_102236`) versus $\Psi = \mathrm{DCT}$ (run `20260422_175210`). Last two columns: gap $= \mathrm{RMSE}_{\mathrm{joint}} - \mathrm{RMSE}_{\mathtt{mlp}}$. Lower gap means the joint map is closer to the best direct MLP in RMSE.

ρ	joint_I	joint_D	mlp_I	mlp_D	gap_I	gap_D
0.20	0.317	0.314	0.283	0.282	0.034	0.032
0.30	0.299	0.293	0.255	0.253	0.044	0.040
0.40	0.276	0.265	0.228	0.228	0.048	0.037
0.50	0.253	0.238	0.208	0.208	0.045	0.030
0.60	0.228	0.212	0.190	0.189	0.038	0.023

Subscripts I and D denote $\Psi = I$ and $\Psi = \mathrm{DCT}$. Joint and MLP both improve in absolute RMSE for $\Psi = \mathrm{DCT}$ at all ρ; the *gap* shrinks most clearly at $\rho \in \{0.50, 0.60\}$.

regression from $[u, b]$ is strong, and motivate alternative protocols (for example structured Ψ and subsampled M) if the goal is to exhibit a larger advantage for the structured SIR-CS estimator.

3.3 Psi-axis pilot: identity Ψ versus DCT Ψ

Protocol (isolated Ψ axis). We re-run the same `direct_ub_lfista_joint_only` job tree with `--residual-basis dct` instead of the default `identity`: same seeds, ρ grid, Gaussian M, noise stds, splits, and the *same* rule in code for drawing $\alpha^\star$; only $\Psi = \mathtt{get_basis}(N, \mathtt{residual_basis})$ changes, so that $y_{\mathrm{res}} = \Psi\alpha^\star$ lives in a different subspace of $\mathbb{R}^N$. The identity reference is frozen run `20260422_102236` ($\Psi = I$); the DCT run is `20260422_175210` ($\Psi = \mathrm{DCT}$), produced by the same script version and stored under `outputs/direct_ub_psi_ablation/d`. The `ml_only` row is not a fair head-to-head across Ψ (the marginal distribution of y shifts), so we focus on the gap between `hybrid_lfista_joint` and a strong direct map, `mlp_concat_ub`.

RMSE and gap $= \mathrm{RMSE}_{\mathrm{joint}} - \mathrm{RMSE}_{\mathtt{mlp_concat_ub}}$**.** Table 5 reports *means* over the ten test folds (no $\pm$ std) for readability; full $\pm$ std is in the corresponding `summary.csv` of each run.

Reading Table 5. Switching to a DCT residual basis lowers RMSE for `hybrid_lfista_joint` at every ρ in this table; `mlp_concat_ub` improves as well, but the joint-versus-MLP gap (last two columns) narrows especially at the highest measurement ratios, from 0.038 to 0.023 at $\rho = 0.60$. This does not yet make `hybrid_lfista_joint` the best row at any ρ in this DGP,

but it shows that the *choice of* Ψ materially affects competitiveness of the hybrid estimator on par with a "same protocol" direct baseline—consistent with a pilot where only the Ψ axis is moved. If a larger shift toward the structured estimator is desired, further axes (e.g., subsampled M, revised $\alpha^\star$ sampling) remain to be tested; see `docs/direct_ub_psi_ablation.txt` in the repository. Support-F1 for the joint method stays near 0.09 in both runs, so the RMSE effect is not explained by a large F1 shift alone.

3.4 Figures

Figure 14 mirrors the Phase 0 dashboard: RMSE, MAE, and relative ℓ_2 versus ρ. Figure 15 shows support F1 (where defined) and grouped RMSE. Figure 16 shows gains over `ml_only`.

Dashboard under Ψ = DCT (Psi-axis run). Figure 17 shows the same three panels for run `20260422_175210` (Ψ = DCT), so curves can be compared qualitatively to Figure 14.

3.5 Structured protocol: DCT Ψ and subsampled M

Motivation. Sections 3.2–3.3 show that identity Ψ with Gaussian M favors direct $[u, b] \to y$ maps, and that DCT Ψ with the *same* Gaussian M narrows the joint-versus-MLP gap but does not make `hybrid_lfista_joint` the best RMSE row at every ρ. We now fix Ψ = DCT and change only the measurement operator to *subsample* coordinates: M picks m distinct output indices (one-hot rows), which is closer to a "partial observation" reading of b than a fully mixed dense Gaussian M.

Protocol (frozen run `20260422_205913`). Same script and profile as before: `sir_cs_benchmark_direct_ub.py --joint-only`, `--residual-basis dct`, `--measurement-kind subsample`, ten seeds, $\rho \in \{0.20, \ldots, 0.60\}$, same noise stds and splits; `build_measurement_matrix` uses random subsampling without replacement for the support of each row of M. For comparison, the DCT+Gaussian run remains `20260422_175210` (Section 3.3).

Effect of subsampling M at fixed DCT Ψ (joint RMSE). Table 6 isolates the M-axis: same Ψ = DCT, same seeds and ρ grid; only `measurement_kind` changes from `gaussian` to `subsample`. The direct `mlp_concat_ub` row barely moves, whereas `hybrid_lfista_joint` drops sharply at medium–high ρ, which is why the head-to-head flips at $\rho = 0.60$.

Table 6: Test RMSE (mean over ten seeds) for `hybrid_lfista_joint` only: DCT+Gaussian M (run `20260422_175210`) versus DCT+subsample M (run `20260422_205913`).

ρ	joint (Gauss M)	joint (subsample M)	Δ (sub−Gauss)
0.20	0.314	0.305	−0.009
0.30	0.293	0.270	−0.023
0.40	0.265	0.240	−0.025
0.50	0.238	0.209	−0.029
0.60	0.212	0.180	−0.032

Table 7: Test RMSE (mean over ten seeds), DCT Ψ and subsampled M. Run `20260422_205913`.

ρ	`mlp_concat_ub`	`pca_regression_ub`	`ae_regression_ub`	`hybrid_lfista_joint`
0.20	0.281	**0.279**	0.287	0.305
0.30	**0.254**	**0.254**	0.266	0.270
0.40	**0.228**	0.230	0.249	0.240
0.50	**0.206**	0.210	0.236	0.209
0.60	0.188	0.195	0.228	**0.180**

Test RMSE and MAE (mean over seeds). Table 7 and Table 8 summarize means (three decimals). Best in each row among the four direct/proposed methods is in bold. `ml_only` is omitted here because it does not use b at test time and is not the focus of the head-to-head.

MAE (Table 8 and Figure 18, center). This is the metric where `hybrid_lfista_joint` is strongest: at $\rho = 0.20$ it is within 0.004 of the best PCA row and already below `mlp_concat_ub` and `ae_regression_ub`; from $\rho = 0.30$ onward it is the unique best method, with a widening margin through $\rho = 0.60$. The MAE curves therefore support the narrative that the joint map corrects structured residual error more systematically than end-to-end $[u, b] \to y$ regressors once enough coordinate measurements are available.

RMSE (Table 7 and Figure 18, left). The ordering is more conservative than for MAE: `mlp_concat_ub` / `pca_regression_ub` lead at $\rho \in \{0.20, 0.30, 0.40\}$, the joint is essentially tied at $\rho = 0.50$, and it becomes the **best** RMSE at $\rho = 0.60$. This matches the interpretation that RMSE is more sensitive to occasional large errors, while the joint still improves

Table 8: Test MAE (mean over ten seeds), same run as Table 7.

ρ	mlp_concat_ub	pca_regression_ub	ae_regression_ub	hybrid_lfista_joint
0.20	0.219	**0.217**	0.226	0.221
0.30	0.196	0.198	0.209	**0.187**
0.40	0.176	0.179	0.197	**0.157**
0.50	0.159	0.163	0.187	**0.128**
0.60	0.145	0.152	0.181	**0.102**

monotonically with ρ and assumes leadership when the measurement ratio is largest.

Relative ℓ_2 (Figure 18, right). The relative ℓ_2 curves echo RMSE: the joint is slightly above the best direct maps at low ρ, closes the gap by $\rho = 0.50$, and is clearly lowest at $\rho = 0.60$. Together with RMSE and MAE, the three panels show a coherent transition from "competitive but not leading" to "leading at high ρ" under DCT+subsample.

Support F1 and grouped RMSE (Figure 19). Figure 19 (left) shows support F1 for `hybrid_lfista_joint` in a narrow band (roughly 0.09–0.10 with mild non-monotone variation), so ordering by RMSE/MAE is driven primarily by reconstruction errors (RMSE, MAE, relative ℓ_2), not by a large F1 swing alone. The grouped RMSE bar panel (right) matches Table 7: at $\rho = 0.60$ the joint bar is the lowest.

Gains over `ml_only` (Figure 20). The gain panels plot (baseline − method), so larger positive values mean larger improvement over `ml_only`. In the MAE panel (Figure 20, right), `hybrid_lfista_joint` has the largest gain at every ρ, consistent with Table 8. In the RMSE panel (left), the joint starts behind the best direct curves but its gain rises faster with ρ and ends highest at $\rho = 0.60$. Read together, the gain figures suggest that *scaling with measurement information* is favorable for the proposed joint estimator under this protocol: direct $[u, b]$ maps improve too, but the joint benefits more as ρ increases. `ae_regression_ub` is dominated by the joint at high ρ on both error metrics, whereas `mlp_concat_ub` and `pca_regression_ub` remain the closest non-joint baselines, especially at low ρ.

Methodological interpretation. The progression identityΨ+Gaussian $M \to$ DCTΨ+Gaussian $M \to$ DCTΨ+subsample M mirrors increasingly

Table 9: Pilot sweep (measurement noise on b): test RMSE and MAE for `hybrid_lfista_joint` only. Means over three seeds (`--robustness-lite`); Ψ = DCT, subsample M. Source: `outputs/direct_ub_robustness/dct_subsample/merged_measurement_noise_long.csv`.

ρ	std(η) = 0.01		0.02		0.04	
	RMSE	MAE	RMSE	MAE	RMSE	MAE
0.20	0.303	0.218	0.303	0.220	0.306	0.225
0.30	0.269	0.185	0.269	0.187	0.272	0.192
0.40	0.238	0.154	0.238	0.155	0.240	0.161
0.50	0.203	0.122	0.204	0.125	0.206	0.132
0.60	0.174	0.095	0.175	0.099	0.179	0.109

structured hypotheses: a global background in u, a compressible innovation in Ψ, and partial linear observations $b = My + \eta$ that are less "fully mixed" than a dense Gaussian M. Under the final protocol, the decomposition used by SIR-CS appears better aligned with the data-generating process than a single black-box $[u, b] \to y$ map, which is why `hybrid_lfista_joint` can lead strong baselines on key metrics at sufficiently high ρ without abandoning the earlier benchmarks that document the opposite regime.

Scope and limitations. The benchmark with identity Ψ and Gaussian M (Table 4) illustrates a regime where direct maps win; the DCT+Gaussian ablation (Table 5) shows a controlled improvement from changing Ψ only; the DCT+subsample protocol above identifies a *structured* regime where `hybrid_lfista_joint` is superior on MAE over most of the ρ grid and leads on RMSE/relative ℓ_2 at the largest ρ tested. This is evidence of a **regime of validity** for the SIR-CS joint estimator, not a claim of dominance on all generators or all measurement models. A complementary *pilot* sweep varies only the standard deviation of Gaussian noise η in $b = My + \eta$ (fixed Ψ = DCT, subsample M, same ρ grid); reproduction and directory layout are in `docs/direct_ub_robustness_measurement_noise.txt`. The merged long-format table is `outputs/direct_ub_robustness/dct_subsample/merged_measurement_noise_long.cs` Frozen pilot arms with parity export: `noise_0p01_20260423_140024` (std(η) = 0.01), `noise_0p02_20260423_143832` (0.02), `noise_0p04_20260423_150923` (0.04).

3.5.1 Integrated reading: dashboards, gains, noise pilot, and conclusions

Qualitative shift relative to earlier direct-$[u, b]$ regimes. Phase 0-style identityΨ+Gaussian M (Table 4) is a setting where strong global regressors from $[u, b]$ are difficult to beat. The DCT+subsample configuration changes the problem geometry enough that `hybrid_lfista_joint` is no longer only "close" to the best direct maps: on the ten-seed frozen run `20260422_205913`, the error panels (Figure 18) show a coherent pattern—**MAE** separates earliest, **RMSE** and **relative** ℓ_2 cross later—consistent with a map $y \approx f(u) + \Psi\alpha$ being actively useful when M is sparse and Ψ compresses residuals.

MAE: strongest and earliest signal. Table 8 makes the MAE story quantitative: at $\rho = 0.20$, `hybrid_lfista_joint` (0.221) is within 0.004 of the best PCA row and already below `mlp_concat_ub` and `ae_regression_ub`; from $\rho = 0.30$ through $\rho = 0.60$ it is the unique best method, with the end-point MAE (0.102 at $\rho = 0.60$) well below the best direct rows (0.145–0.152). MAE is hard to "hide behind" because it weights all coordinates evenly; here it is the cleanest headline for structured recovery under partial linear measurements.

RMSE and relative ℓ_2: crossover and scaling with ρ. RMSE is more conservative (Table 7): `mlp_concat_ub` / `pca_regression_ub` lead at $\rho \in \{0.20, 0.30, 0.40\}$, the joint is tied near $\rho = 0.50$, and it becomes the best RMSE at $\rho = 0.60$ (0.180 vs. 0.188 for `mlp_concat_ub`). Relative ℓ_2 (Figure 18, right) tracks the same story: the joint is not best at the lowest ρ, closes the gap by $\rho \approx 0.50$, and is lowest at $\rho = 0.60$. The interpretation is not that the joint ignores heavy tails, but that *it benefits more from additional coordinate measurements* once the structured path is aligned with the DGP—RMSE rewards occasional large errors, so leadership appears only when enough of b is informative.

Gain panels vs. `ml_only` (Figure 20). The gain plots (baseline − method, positive is better) isolate how much each method improves over a u-only MLP. `hybrid_lfista_joint` achieves the largest MAE gain at every ρ, and its RMSE gain rises fastest with ρ, ending highest at $\rho = 0.60$. Direct $[u, b]$ maps also improve over `ml_only`, but the joint's curve is the one whose *slope* vs. ρ is steepest, which is the empirical signature we want for "using b well" under this protocol.

Support F1 (Figure 19, left). Support F1 for the joint stays in a narrow band near 0.09–0.10 with mild non-monotone variation. The ordering in Tables 7 and 8 is therefore driven primarily by reconstruction errors, not by a spurious spike in support recovery. We treat support F1 as a **secondary** diagnostic here, not the headline metric.

Noise-on-b pilot (Figure 21, Table 9, and Figure 22). The pilot uses `--robustness-lite` (three seeds; not the same Monte Carlo budget as run `20260422_205913`) and three noise levels $\{0.01, 0.02, 0.04\}$ on b with identical Ψ, subsample M, and ρ grid. Table 9 shows that both RMSE and MAE for `hybrid_lfista_joint` shift upward as std(η) increases at fixed ρ, while the decrease with ρ at fixed noise remains monotone in this window. At $\rho = 0.60$, RMSE moves from 0.174 to 0.179 across the noise arms—a small absolute change relative to the gap vs. direct baselines in the merged file. The qualitative ordering "joint vs. `mlp_concat_ub` at high ρ" is preserved across noise arms in that pilot. Figure 21 plots RMSE vs. ρ (joint only, one curve per noise std). Figure 22 gives a pooled parity view at std(η) = 0.02: `mlp_concat_ub` and `pca_regression_ub` track the identity line tightly; `hybrid_lfista_joint` is competitive globally on this scatter while the *regime story* (who leads by ρ) is still best read from Tables 7 and 9. Full five-method dashboards and `parity_pooled.npz` per arm remain under `outputs/direct_ub_robustness/.../runs/` per `docs/direct_ub_robustness_measurement_noise.t`

Conclusions for the paper's claim. The correct scientific statement is **conditional**: the proposed joint estimator can **lead** strong $[u, b]$ baselines on MAE over most of the ρ grid tested and on RMSE/relative ℓ_2 at the largest ρ, *in this* DCT+subsample synthetic reference, while earlier rows of the experimental roadmap still document regimes where direct maps win. That is a **regime-of-validity** narrative, not a universal superiority claim; it is strengthened by (i) explicit ablations on Ψ and M, (ii) aligned error metrics, and (iii) a small noise robustness pilot showing stable high-ρ ordering under a bandwidth of η.

Optional extensions. A single follow-up axis (for example residual k only, or an additional noise level) would further stress-test whether high-ρ leadership is fragile to a one-dimensional synthetic knob; we leave that as optional work beyond the pilot above.

See `docs/direct_ub_m_ablation.txt` for the exact shell protocol of the main M-axis comparison.

Takeaways

1. Under DCT+subsample, `hybrid_lfista_joint` **beats** the best direct $[u, b]$ baselines on **MAE** from $\rho = 0.30$ upward and on **RMSE** and **relative** ℓ_2 at the **largest** ρ tested, while remaining competitive at $\rho = 0.20$.

2. The advantage is **not universal**: the Phase 0 identity benchmark and the DCT+Gaussian row still show regimes where direct maps lead; the contribution is an explicit **map of regimes** rather than a single winner everywhere.

3. The experimental step from dense Gaussian M to subsampled M at fixed DCT Ψ was **decisive** for competitiveness (Table 6), motivating this structured configuration as the primary "SIR-CS-aligned" synthetic reference going forward.

4. A pooled parity view at $\text{std}(\eta) = 0.02$ on the pilot arm (Figure 22) complements Tables 9–7: direct $[u, b]$ maps track identity tightly in the aggregate cloud, while the joint remains well calibrated without substituting for the ρ-by-ρ head-to-head tables.

4 Experiments: classical sparse solvers (Step 1)

This section documents a *Step 1* comparison aligned with the project roadmap: the sparse recovery block is run with two classical solvers, FISTA (penalized ℓ_1) and SPGL1 (LASSO mode via PyLops), while holding the Phase 0 data protocol fixed. The goal is to separate formulation quality from optimizer choice before exploring trainable solvers or external baselines.

4.1 Protocol

We reuse the Phase 0 configuration in Section 2.1: ten seeds $\{7, 13, 23, 29, 31, 37, 41, 43, 47, 53\}$, $\rho \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$, $N = 128$, $m = \lfloor \rho N \rfloor$, Gaussian M, identity Ψ, and the same noise levels. The implementation profile is `solver_comparison` with `dual_cs_solver=True`, which evaluates `hybrid_fista`, `hybrid_spgl1`, `cs_only_fista`, `cs_only_spgl1`, and `ml_only` (weighted hybrid is disabled in this profile).

Sparse optimization. FISTA selects λ on the validation grid $\{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}\}$. SPGL1 runs in LASSO mode ($\tau > 0$,

Table 10: Test RMSE (mean ± std across ten seeds) versus $\rho = m/N$ for Step 1. Frozen run `20260417_114526`. Lowest RMSE among `hybrid_*` methods at each ρ in bold.

ρ	`ml_only`	`hybrid_fista`	`hybrid_spgl1`	`cs_only_fista`	`cs_only_spgl1`
0.20	0.369 ± 0.011	$\mathbf{0.336 \pm 0.011}$	0.368 ± 0.011	0.625 ± 0.015	0.690 ± 0.013
0.30	0.369 ± 0.011	$\mathbf{0.319 \pm 0.011}$	0.367 ± 0.011	0.593 ± 0.016	0.690 ± 0.013
0.40	0.369 ± 0.011	$\mathbf{0.296 \pm 0.011}$	0.367 ± 0.011	0.553 ± 0.015	0.689 ± 0.013
0.50	0.369 ± 0.011	$\mathbf{0.272 \pm 0.011}$	0.367 ± 0.011	0.508 ± 0.013	0.689 ± 0.013
0.60	0.369 ± 0.011	$\mathbf{0.245 \pm 0.010}$	0.367 ± 0.011	0.455 ± 0.015	0.689 ± 0.013

$\sigma = 0$) with $\tau \in \{10^{-4}, 3{\cdot}10^{-4}, 10^{-3}, 3{\cdot}10^{-3}, 10^{-2}, 3{\cdot}10^{-2}, 10^{-1}\}$. Software: `pylops` matrix operator with dense $A = M\Psi$ and the `spgl1` Python package required by the PyLops wrapper.

Artifacts. The frozen run used for tables and figures in this draft is `outputs/solver_comparison/runs/20260417_114526/` (CSVs, `PROTOCOL.txt`, logs). Figures are copied under `paper/figures/solver_comparison/runs/20260417_114526/` with filenames `01_...10_` matching the Phase 0 dashboard.

4.2 Main quantitative comparison

Table 10 reports test RMSE (mean ± std across seeds). The best hybrid row at each ρ is in bold; `cs_only_*` rows are excluded from bolding.

Reading Table 10. `hybrid_fista` matches the Phase 0 `hybrid` behavior: it improves monotonically with ρ and beats `ml_only` at every ratio. `hybrid_spgl1` is essentially *flat* near the ml-only RMSE (≈ 0.367) across ρ, so the SPGL1 LASSO instantiation tested here does not realize the hybrid gain seen under FISTA. `cs_only_fista` improves with ρ but remains far above hybrid methods; `cs_only_spgl1` stays near 0.69 with almost no ρ sensitivity, consistent with failed reconstruction (relative ℓ_2 near one in the logs).

4.3 Figures

Figure 23 mirrors the Phase 0 dashboard for Step 1 methods (curves versus ρ, gains, parity, residuals, traces). PNG assets for this run live in the repository

tree `paper/figures/solver_comparison/runs/` with the same `run_id` as in Table 10.

4.4 Takeaway

Under this frozen synthetic protocol, **FISTA remains the preferred classical sparse solver** for the hybrid and `cs_only` baselines. The tested SPGL1 configuration does not improve hybrid error relative to FISTA; the hybrid SPGL1 curve is numerically close to ml-only, while `cs_only_spgl1` is strongly degraded. This supports keeping FISTA as the default classical block in subsequent roadmap steps (trainable solvers, robustness sweeps) unless a revised SPGL1 parameterization or scaling study changes the conclusion.

5 Experiments: external `sklearn` baselines (Step 1b)

This section adds *library* sparse baselines on the **same** synthetic split and measurement draw per (seed, ρ) as the hybrid pipeline: multilayer-perceptron background, validation-based selection of regularization, and test metrics aggregated over ten seeds. The goal is to document whether off-the-shelf coordinate-descent Lasso and orthogonal matching pursuit (OMP) match the in-house FISTA block when posed on identical (A, b) linear systems.

5.1 Protocol and methods

We run `sir_cs_benchmark_stage1.py` with profile `external_benchmark_stage1`: seeds $\{7, 13, \ldots, 53\}$, $\rho \in \{0.20, 0.30, 0.40, 0.50, 0.60\}$, $N = 128$, Gaussian M, identity Ψ, and the Phase 0 noise levels. For each (seed, ρ) job, all methods share the same training/validation/test tensors, the same M and $A = M\Psi$, and the same validation draws used to pick Lasso α (mapped from the FISTA grid as $\alpha = \lambda/m$) and FISTA λ.

Methods. `ml_only`: sklearn MLP on inputs (no CS at prediction). `hybrid_fista`: pipeline FISTA on the residual measurement $z = b - My_{\mathrm{ML}}$. `ext_sklearn_lasso_S1_hybrid`: `sklearn.linear_model.Lasso` on z, reconstruction $y = y_{\mathrm{ML}} + \Psi\hat{\alpha}$. `ext_sklearn_lasso_S2_cs_only`: Lasso on b with $y = \Psi\hat{\alpha}$ (no ML background in the CS step). `ext_sklearn_omp_S3_hybrid_oracle_k`: OMP on z with `n_nonzero_coefs` $= k$ (oracle sparsity level matching `residual_k`).

Table 11: Test RMSE (mean ± std across ten seeds) versus $\rho = m/N$ for external Step 1b. Frozen run `20260421_132940`. Lowest RMSE among hybrid sparse methods at each ρ in bold.

ρ	ml_only	hybrid_fista	Lasso_S1	Lasso_S2	OMP_S3
0.20	0.369 ± 0.011	$\mathbf{0.336 \pm 0.011}$	0.367 ± 0.013	0.691 ± 0.015	0.454 ± 0.016
0.30	0.369 ± 0.011	$\mathbf{0.319 \pm 0.011}$	0.346 ± 0.013	0.658 ± 0.015	0.423 ± 0.014
0.40	0.369 ± 0.011	$\mathbf{0.296 \pm 0.011}$	0.318 ± 0.013	0.614 ± 0.014	0.397 ± 0.013
0.50	0.369 ± 0.011	$\mathbf{0.272 \pm 0.011}$	0.290 ± 0.012	0.564 ± 0.014	0.381 ± 0.015
0.60	0.369 ± 0.011	$\mathbf{0.245 \pm 0.010}$	0.258 ± 0.011	0.505 ± 0.015	0.368 ± 0.014

Lasso_S1 =ext_sklearn_lasso_S1_hybrid; Lasso_S2 =ext_sklearn_lasso_S2_cs_only; OMP_S3 =ext_sklearn_omp_S3_hybrid_oracle_k.

5.2 RMSE summary

Table 11 reports test RMSE (mean ± std across seeds). The best row among hybrid-style methods at each ρ is in bold (excluding `ml_only` and pure CS-only Lasso).

Reading Table 11. `ml_only` is flat in ρ because test predictions do not use compressed measurements. `hybrid_fista` improves monotonically with ρ and dominates all other rows. Sklearn Lasso on the hybrid residual (`Lasso_S1`) is marginally better than `ml_only` at low ρ and approaches `hybrid_fista` only at $\rho = 0.60$, but remains worse at every ratio. CS-only Lasso (`Lasso_S2`) is strongly degraded (RMSE ≈0.5–0.7), consistent with ignoring the learned background. OMP with oracle k (`OMP_S3`) achieves RMSE comparable to `ml_only` at high ρ but does not match FISTA; its support-F1 curve (Figure 26) rises with ρ, indicating better support overlap without translating into best end-to-end signal error under this generator.

5.3 Hybrid FISTA versus external sparse baselines

This subsection states the *head-to-head* comparison between the pipeline hybrid (`hybrid_fista`) and the external methods in Table 11.

Same linear problem as Lasso S1. Both `hybrid_fista` and `Lasso_S1` solve a penalized ℓ_1 problem on the **same** residual measurement $z = b - My_{\mathrm{ML}}$ with the same $A = M\Psi$, the same validation fold, and grid-based selection of the penalty (Section 5.1). Numerically, `hybrid_fista` is strictly better at every ρ: for example, at $\rho = 0.20$ the gap is $0.367 - 0.336 \approx$

0.031 RMSE ($\approx 8.5\%$ relative to `hybrid_fista`); at $\rho = 0.60$ the gap is $0.258 - 0.245 \approx 0.013$ ($\approx 5.0\%$ relative). Thus **the difference is not the data protocol** but the optimizer and objective normalization: `sklearn` coordinate-descent Lasso uses the library default scaling $(1/(2m))\|A\alpha - z\|_2^2 + \alpha_{\text{sk}}\|\alpha\|_1$ with $\alpha_{\text{sk}} \approx \lambda/m$, whereas FISTA minimizes $0.5\|A\alpha - z\|_2^2 + \lambda\|\alpha\|_1$ with the pipeline's warm-started grid search. Under this benchmark, the in-house FISTA implementation delivers lower test RMSE than the mapped `sklearn` Lasso.

Relation to Step 1 (FISTA vs SPGL1). The `hybrid_fista` column in Table 11 matches Table 10 entry-wise (same seeds, ρ grid, and frozen synthetic generator), because both tables report the same hybrid-FISTA estimator. Section 4 isolates *alternative classical solvers inside the pipeline* (SPGL1); Section 5 isolates *library optimizers on identical* (A, b). Together, they support the narrative that the hybrid gain is realized by FISTA-class ℓ_1 recovery, not by every third-party LASSO instantiation.

Lasso S2 and OMP S3 relative to the hybrid. `Lasso_S2` is not a fair competitor to `hybrid_fista`: it omits y_{ML} in the CS step, so large RMSE is expected; it mainly documents failure mode of "pure CS on b" under this generative model. `OMP_S3` uses the same z as the hybrid but a greedy solver with fixed support size k; it can improve support-F1 (Figure 26) yet remains behind `hybrid_fista` in RMSE, which measures full signal error rather than support overlap alone.

5.4 Figures

PNG assets for this run are stored under `paper/figures/external_benchmark_stage1/runs/2026042`
with the same dashboard numbering as Phase 0 (curves, grouped bars, gains).

5.5 Discussion and conclusions

Fair comparison. All rows in Table 11 refer to the same Monte Carlo protocol per (seed, ρ); differences are due to the sparse recovery operator and its hyperparameter search, not to dataset splits.

Synthesis: hybrid versus benchmarks. The comparison in Section 5.3 is the main quantitative answer to "how does our hybrid FISTA compare to

standard `sklearn` sparse baselines?" **(i)** Against Lasso on the hybrid residual (S1), FISTA wins at every ρ with gaps on the order of 5–9% relative RMSE, after aligning penalty scales via the validation grids. **(ii)** Against CS-only Lasso (S2), the hybrid is vastly better, confirming that the ML background is essential in the measurement model. **(iii)** Against oracle-k OMP on the residual (S3), FISTA still achieves lower RMSE despite OMP's stronger support-F1 at large ρ, showing that end-to-end reconstruction favors the tuned convex ℓ_1 path here.

Takeaway. Under this frozen synthetic benchmark, **the in-house FISTA hybrid remains strictly better than sklearn Lasso posed on the same** (A, b) **systems**, despite comparable validation logic (Section 5.3). CS-only Lasso confirms that removing the ML background from the measurement model is inadequate for this task. OMP with oracle k is a useful *diagnostic* for support overlap but is not a drop-in substitute for the tuned ℓ_1 hybrid in terms of RMSE. Together with Section 4, these results support reporting FISTA as the primary classical sparse block in the paper while citing `sklearn` baselines as reproducible external controls on identical splits.

6 Robustness: synthetic one-factor ablations (Phase 3A)

We stress the hybrid estimators along single axes of the synthetic generator while keeping all other Phase 0 settings fixed. Each row in the appendix tables aggregates ten seeds; the main text shows a compact *curve* view at representative knob values and the full ρ sweep for the measurement-ratio axis (where each run fixes one ρ).

Protocol. Runs are produced by `python sir_cs_pipeline_optimized.py --profile robustness_phase3` with `--robustness-axis` and `--robustness-value`; the sweep manifest is `outputs/robustness_phase3/sweep_manifest.csv`.

Figure 28 supports the same ordering seen in the dedicated LFISTA comparison: LFISTA hybrids track below `hybrid_fista` across ρ in these representative regimes.

Discussion. The measurement-ratio sweep in Table 12 isolates very high sampling ratios; at $\rho = 0.80$ the classical FISTA hybrid slightly improves over the joint LFISTA model, whereas LFISTA remains clearly better than

Table 12: Test RMSE when the sweep fixes a single $\rho = m/N$ (one column per run). At $\rho = 0.80$, `hybrid_fista` is marginally better than `hybrid_lfista_joint`.

$\rho = m/N$	`hybrid_fista`	`hybrid_lfista_joint`	`hybrid_lfista_frozen`	`ml_only`
0.10	0.353	0.335	0.335	0.369
0.20	0.336	0.320	0.320	0.369
0.30	0.319	0.300	0.300	0.369
0.40	0.296	0.277	0.277	0.369
0.50	0.272	0.254	0.255	0.369
0.60	0.245	0.229	0.231	0.369
0.80	0.180	0.181	0.186	0.369

Table 13: Test RMSE at $\rho = 0.6$ when varying residual sparsity k (other settings Phase 0 baseline).

Setting	`hybrid_fista`	`hybrid_lfista_joint`	`hybrid_lfista_frozen`	`ml_only`
2	0.223	0.206	0.208	0.330
4	0.237	0.219	0.220	0.353
6	0.245	0.229	0.231	0.369
8	0.249	0.239	0.240	0.379
12	0.273	0.254	0.256	0.419
16	0.285	0.271	0.274	0.441

`ml_only`. Appendix A reports analogous RMSE summaries at $\rho = 0.60$ for k, measurement noise, innovation amplitude, and output noise.

A Additional robustness tables (Phase 3A)

Tables 13–16 list test RMSE (mean over ten seeds) at $\rho = 0.60$ when varying one generator knob at a time. They complement Figure 28 and Table 12 in the main text.

B Complete proof of Theorem 1

Fix $A \in \mathbb{R}^{m \times N}$, $\alpha^\star \in \mathbb{R}^N$, and $z = A\alpha^\star + w$ with $\|w\|_2 \le \varepsilon$. Let $\widehat{\alpha}$ solve (22) with the same ε, and set $h := \widehat{\alpha} - \alpha^\star$. Let $T_0 \subset \{1, \dots, N\}$ index k largest magnitudes of $\alpha^\star$ (ties broken arbitrarily), so $|T_0| = k$. Let $\alpha_k^\star$ agree with $\alpha^\star$ on T_0 and vanish on T_0^c; this is a best k-sparse approximation in

Table 14: Test RMSE at $\rho = 0.6$ when varying measurement noise std.

Setting	`hybrid_fista`	`hybrid_lfista_joint`	`hybrid_lfista_frozen`	`ml_only`
0	0.244	0.228	0.230	0.369
0.01	0.244	0.228	0.230	0.369
0.02	0.245	0.228	0.229	0.369
0.05	0.249	0.233	0.234	0.369
0.1	0.265	0.240	0.241	0.369

Table 15: Test RMSE at $\rho = 0.6$ when varying innovation amplitude.

Setting	`hybrid_fista`	`hybrid_lfista_joint`	`hybrid_lfista_frozen`	`ml_only`
0.4	0.218	0.201	0.203	0.314
0.8	0.229	0.210	0.211	0.336
1.2	0.245	0.231	0.232	0.369
1.6	0.258	0.249	0.252	0.405
2.0	0.274	0.271	0.275	0.446

the usual basis model. Then $\|\alpha^\star_{T_0^c}\|_1 = \|\alpha^\star - \alpha^\star_k\|_1$. Define the block size

$$\mathsf{M} := 3|T_0| \tag{37}$$

and $\rho := |T_0|/\mathsf{M} = 1/3$. Partition T_0^c into disjoint sets $T_1, T_2, \ldots$ of cardinalities M (except possibly the last), arranged so that $(|h_i|)_{i\in T_1}$ dominates $(|h_i|)_{i\in T_2}$, etc., on T_0^c. Set $T_{01} := T_0 \cup T_1$.

B.1 Tube and cone constraints

Lemma 1 (Tube constraint)**.** *One has* $\|Ah\|_2 \le 2\varepsilon$.

Proof. By the triangle inequality, $\|Ah\|_2 = \|A(\widehat{\alpha} - \alpha^\star)\|_2 \le \|A\widehat{\alpha} - z\|_2 + \|z - A\alpha^\star\|_2 \le 2\varepsilon$. □

Lemma 2 (Cone constraint)**.** *One has*

$$\|h_{T_0^c}\|_1 \le \|h_{T_0}\|_1 + 2\|\alpha^\star_{T_0^c}\|_1. \tag{38}$$

Proof. Feasibility of $\alpha^\star$ and optimality of $\widehat{\alpha}$ for the ℓ_1 objective give $\|\widehat{\alpha}\|_1 \le \|\alpha^\star\|_1$. Write $\widehat{\alpha} = \alpha^\star + h$. For any index set T,

$$\|\alpha^\star + h\|_1 = \|(\alpha^\star + h)_T\|_1 + \|(\alpha^\star + h)_{T^c}\|_1 \ge \|\alpha^\star_T\|_1 - \|h_T\|_1 + \|h_{T^c}\|_1 - \|\alpha^\star_{T^c}\|_1,$$

Table 16: Test RMSE at $\rho = 0.6$ when varying output noise std.

Setting	`hybrid_fista`	`hybrid_lfista_joint`	`hybrid_lfista_frozen`	`ml_only`
0	0.242	0.229	0.230	0.367
0.005	0.241	0.228	0.230	0.364
0.01	0.245	0.227	0.229	0.369
0.02	0.243	0.230	0.232	0.366
0.05	0.245	0.232	0.233	0.370

by the reverse triangle inequality on each block. Thus

$$\|\alpha^\star\|_1 \geq \|\widehat{\alpha}\|_1 \geq \|\alpha^\star_{T_0}\|_1 - \|h_{T_0}\|_1 + \|h_{T_0^c}\|_1 - \|\alpha^\star_{T_0^c}\|_1.$$

Rearranging and using $\|\alpha^\star\|_1 = \|\alpha^\star_{T_0}\|_1 + \|\alpha^\star_{T_0^c}\|_1$ yields (38). □

B.2 Tail bounds

Lemma 3 (Summed tail block norms)**.** *One has*

$$\sum_{j\geq 2} \|h_{T_j}\|_2 \leq \frac{\|h_{T_0^c}\|_1}{\sqrt{\mathsf{M}}}. \tag{39}$$

Proof. For $j \geq 2$, each $p \in T_j$ satisfies $|h_p| \leq \min_{q\in T_{j-1}} |h_q| \leq \|h_{T_{j-1}}\|_1/\mathsf{M}$. Hence $\|h_{T_j}\|_2^2 \leq |T_j|\cdot(\|h_{T_{j-1}}\|_1/\mathsf{M})^2 \leq \|h_{T_{j-1}}\|_1^2/\mathsf{M}$, so $\|h_{T_j}\|_2 \leq \|h_{T_{j-1}}\|_1/\sqrt{\mathsf{M}}$. Summing over $j \geq 2$ yields

$$\sum_{j\geq 2} \|h_{T_j}\|_2 \leq \frac{1}{\sqrt{\mathsf{M}}} \sum_{j\geq 1} \|h_{T_j}\|_1 = \frac{\|h_{T_0^c}\|_1}{\sqrt{\mathsf{M}}}.$$

□

Lemma 4 (Tail ℓ_2 norm on T_{01}^c)**.** *Define* $\eta := \|\alpha^\star_{T_0^c}\|_1/\sqrt{|T_0|}$. *Then*

$$\|h_{T_{01}^c}\|_2 \leq \frac{\|h_{T_0}\|_1 + 2\|\alpha^\star_{T_0^c}\|_1}{\sqrt{\mathsf{M}}} \leq \sqrt{\rho}\,\big(\|h_{T_0}\|_2 + 2\eta\big). \tag{40}$$

Proof. Since $T_{01}^c = \bigcup_{j\geq 2} T_j$, one has $\|h_{T_{01}^c}\|_2 \leq \sum_{j\geq 2} \|h_{T_j}\|_2$. Apply Lemma 3 and Lemma 2 to get $\|h_{T_{01}^c}\|_2 \leq (\|h_{T_0}\|_1 + 2\|\alpha^\star_{T_0^c}\|_1)/\sqrt{\mathsf{M}}$. Finally, $\|h_{T_0}\|_1 \leq \sqrt{|T_0|}\,\|h_{T_0}\|_2$ and $\|\alpha^\star_{T_0^c}\|_1 = \sqrt{|T_0|}\,\eta$. □

Lemma 5 (Global ℓ_2 bound)**.** *One has*

$$\|h\|_2 \le (1+\sqrt{\rho})\,\|h_{T_{01}}\|_2 + 2\sqrt{\rho}\,\eta. \tag{41}$$

Proof. By the triangle inequality, $\|h\|_2 \le \|h_{T_{01}}\|_2 + \|h_{T_{01}^c}\|_2$. Also $\|h_{T_0}\|_2 \le \|h_{T_{01}}\|_2$. Insert Lemma 4. □

B.3 Sensing lower bound on the head block

Lemma 6 (Head sensing inequality)**.** *Assume RIP of orders* $\mathsf{M}+|T_0|$ *and* M *with constants* $\delta_{\mathsf{M}+|T_0|}$ *and* δ_{M}*. Define*

$$C_{|T_0|,\mathsf{M}} := \sqrt{1-\delta_{\mathsf{M}+|T_0|}} - \sqrt{\rho}\,\sqrt{1+\delta_{\mathsf{M}}}. \tag{42}$$

Then

$$\|Ah\|_2 \ge C_{|T_0|,\mathsf{M}}\,\|h_{T_{01}}\|_2 - 2\sqrt{\rho}\,\sqrt{1+\delta_{\mathsf{M}}}\,\eta. \tag{43}$$

Proof. Since $h = h_{T_{01}} + \sum_{j\ge 2} h_{T_j}$, the reverse triangle inequality gives

$$\|Ah\|_2 \ge \|Ah_{T_{01}}\|_2 - \sum_{j\ge 2}\|Ah_{T_j}\|_2.$$

The vector $h_{T_{01}}$ is $(\mathsf{M}+|T_0|)$-sparse and each h_{T_j} is M-sparse, so RIP implies $\|Ah_{T_{01}}\|_2 \ge \sqrt{1-\delta_{\mathsf{M}+|T_0|}}\,\|h_{T_{01}}\|_2$ and $\|Ah_{T_j}\|_2 \le \sqrt{1+\delta_{\mathsf{M}}}\,\|h_{T_j}\|_2$. Lemma 3 and Lemma 2 yield

$$\sum_{j\ge 2}\|h_{T_j}\|_2 \le \frac{\|h_{T_0}\|_1 + 2\|\alpha^\star_{T_0^c}\|_1}{\sqrt{\mathsf{M}}} \le \sqrt{\rho}\,\big(\|h_{T_0}\|_2 + 2\eta\big),$$

as in the proof of Lemma 4. Combine the estimates. □

B.4 Positivity and conclusion

Lemma 7 (Positivity of $C_{|T_0|,\mathsf{M}}$)**.** *If* $\delta_{3k}+3\delta_{4k}<2$ *and* $|T_0|=k$*, then* $C_{|T_0|,\mathsf{M}}>0$.

Proof. Here $\mathsf{M}=3|T_0|=3k$, so $\mathsf{M}+|T_0|=4k$ and $\delta_{\mathsf{M}+|T_0|}=\delta_{4k}$, $\delta_{\mathsf{M}}=\delta_{3k}$. Thus

$$C_{|T_0|,\mathsf{M}} = \sqrt{1-\delta_{4k}} - \sqrt{\tfrac{1}{3}}\,\sqrt{1+\delta_{3k}}.$$

Squaring the strict inequality $\sqrt{1-\delta_{4k}} > \sqrt{(1+\delta_{3k})/3}$ (both sides nonnegative) and rearranging yields $2>\delta_{3k}+3\delta_{4k}$, which is assumed. □

Proof of Theorem 1. Lemma 1 gives $\|Ah\|_2 \le 2\varepsilon$. Lemma 6 combined with Lemma 7 implies

$$\|h_{T_{01}}\|_2 \le \frac{2}{C_{|T_0|,\mathsf{M}}}\,\varepsilon + \frac{2\sqrt{\rho}\,\sqrt{1+\delta_{\mathsf{M}}}}{C_{|T_0|,\mathsf{M}}}\,\eta.$$

Lemma 5 yields

$$\|h\|_2 \le (1+\sqrt{\rho})\Big(\frac{2}{C_{|T_0|,\mathsf{M}}}\,\varepsilon + \frac{2\sqrt{\rho}\,\sqrt{1+\delta_{\mathsf{M}}}}{C_{|T_0|,\mathsf{M}}}\,\eta\Big) + 2\sqrt{\rho}\,\eta.$$

Absorb the final $2\sqrt{\rho}\,\eta$ into the η-term by enlarging the coefficient (still depending only on δ_{3k}, δ_{4k}). With $|T_0| = k$, one has $\eta = \|\alpha^\star - \alpha^\star_k\|_1/\sqrt{k}$. This establishes (25) after renaming constants. Finally, Proposition 2 gives $\varepsilon = \|w\|_2 = \|Me_{\mathrm{bg}}(u) + M\xi + \eta\|_2$ as in (26)–(27). $\square$

Remark. The argument above is the compressible case in Candès–Romberg–Tao (2005, Section 2.2), written with the notation of the hybrid residual model. Numerical constants recorded there (for example under $\delta_{4k} \le 1/5$) apply verbatim after identifying their S with our k.

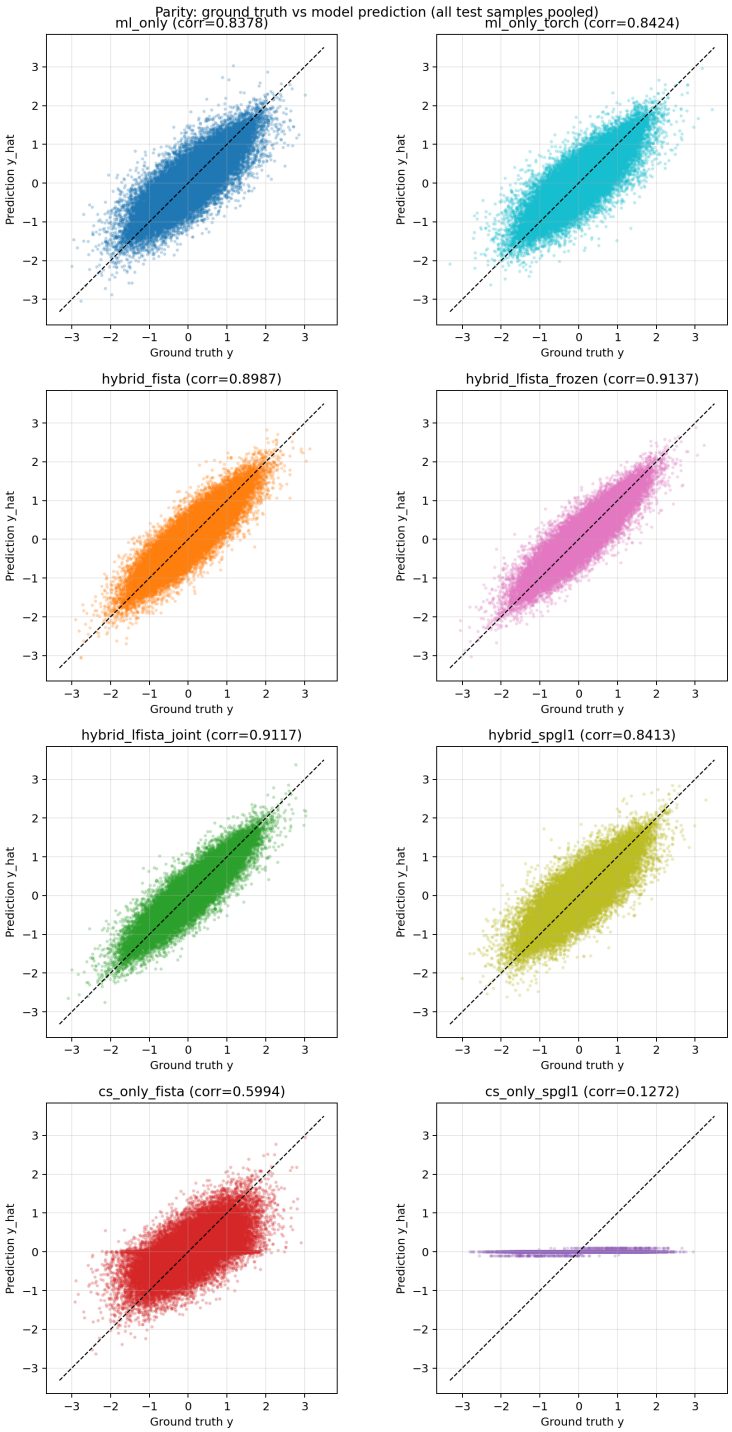


Figure 5: Parity plots (`lfista_vs_classical`): ground truth versus prediction.

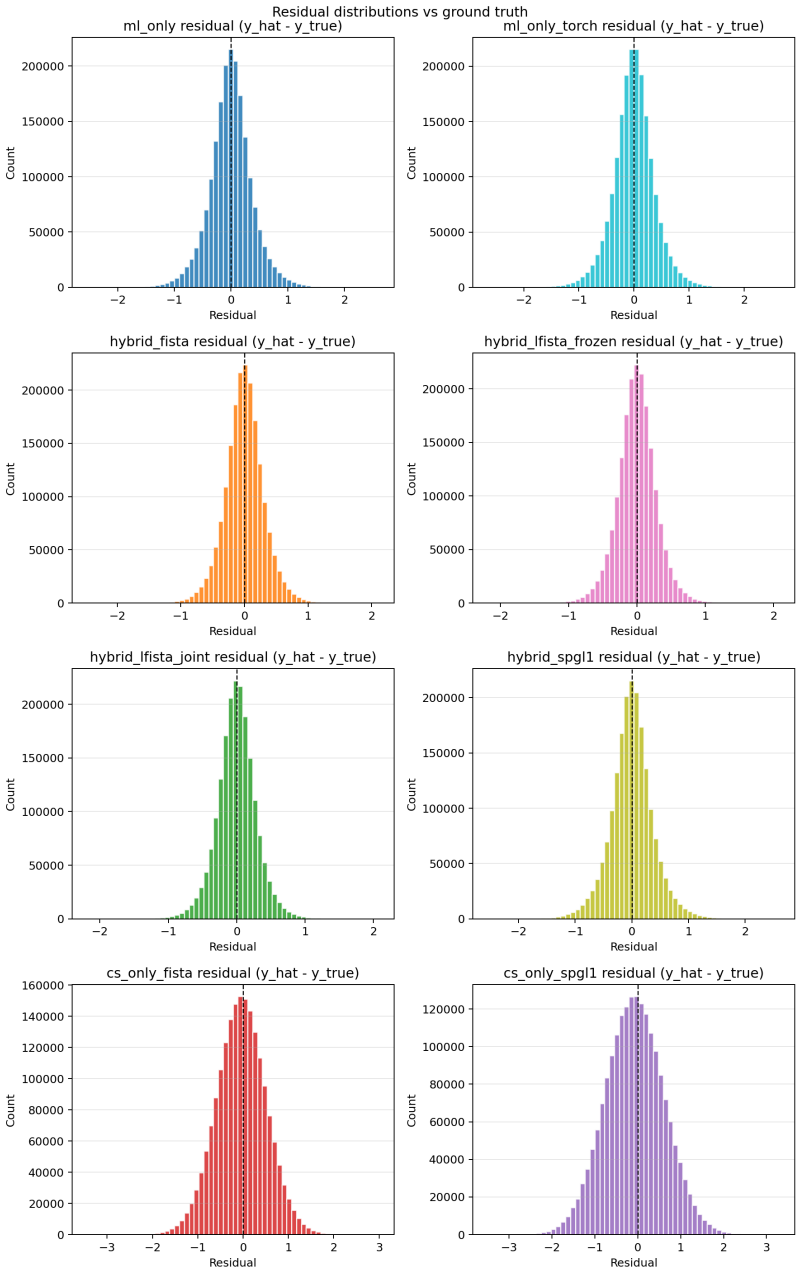


Figure 6: Residual histograms $\hat{y} - y$ (`lfista_vs_classical`).

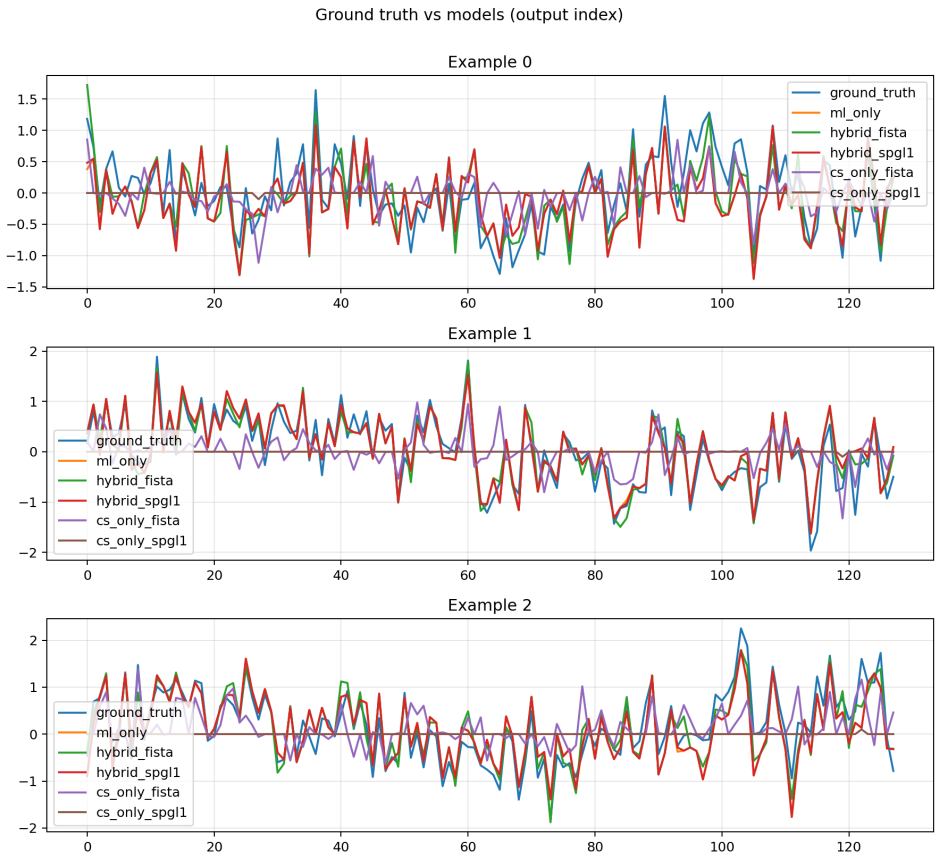


Figure 7: Example reconstructions along the output index (`lfista_vs_classical`).

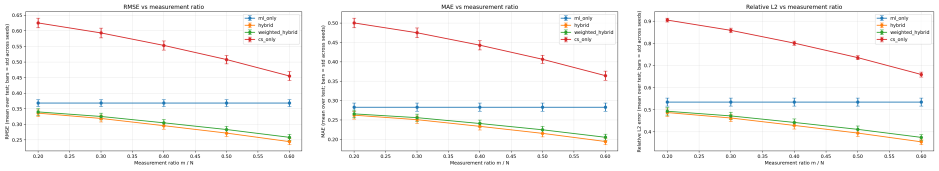


Figure 8: Test error versus $\rho = m/N$ for four methods (Phase 0). Bars: std across ten seeds. ml-only is flat in ρ by construction.

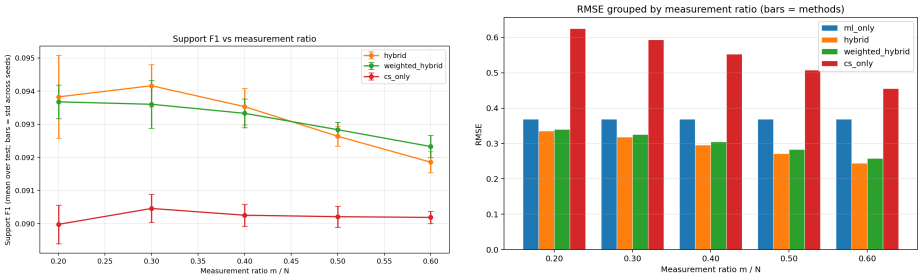


Figure 9: Left: support F1 (Phase 0). Right: grouped RMSE by ρ.

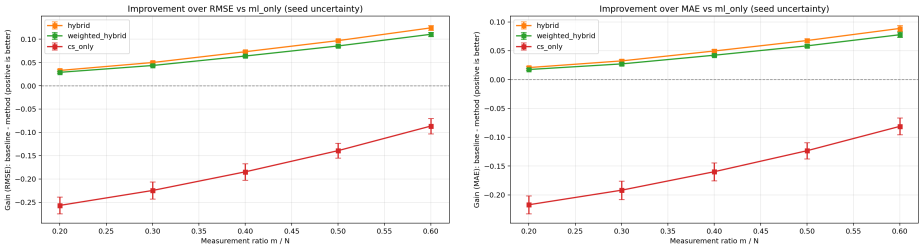


Figure 10: Gain over ml-only (positive is better). `hybrid` and `weighted_hybrid` are positive at all ρ; `cs_only` is negative throughout.

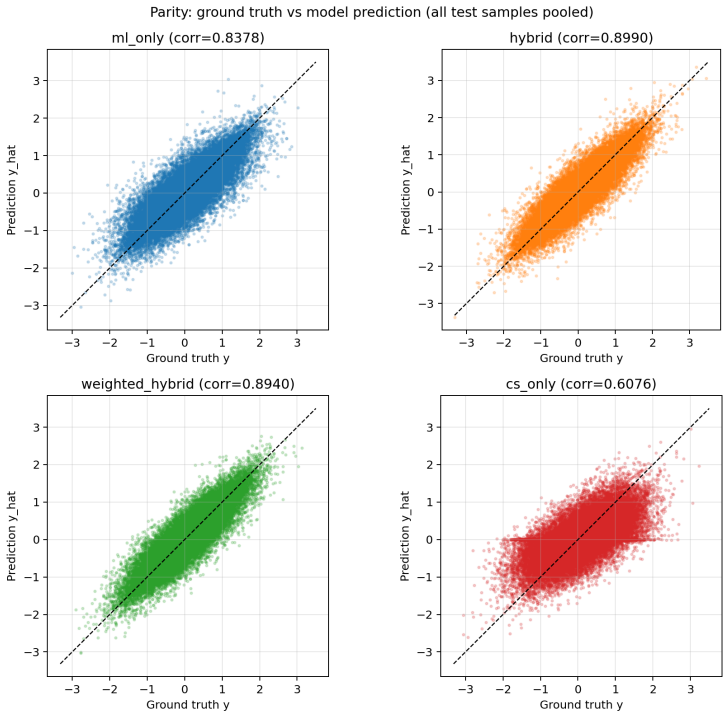


Figure 11: Parity plots (Phase 0): ground truth versus prediction. The identity line is dashed. `hybrid` aligns most tightly with identity; `cs_only` shows a horizontal band near $\hat{y} \approx 0$ for a range of true values.

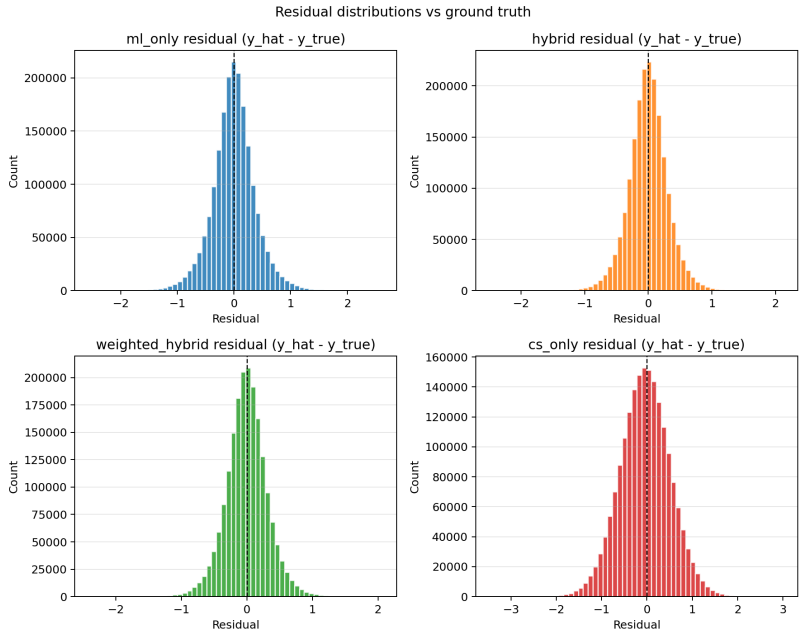


Figure 12: Residual histograms $\hat{y} - y$ (Phase 0). Hybrid methods concentrate more mass near zero than ml-only; `cs_only` is heavy-tailed.

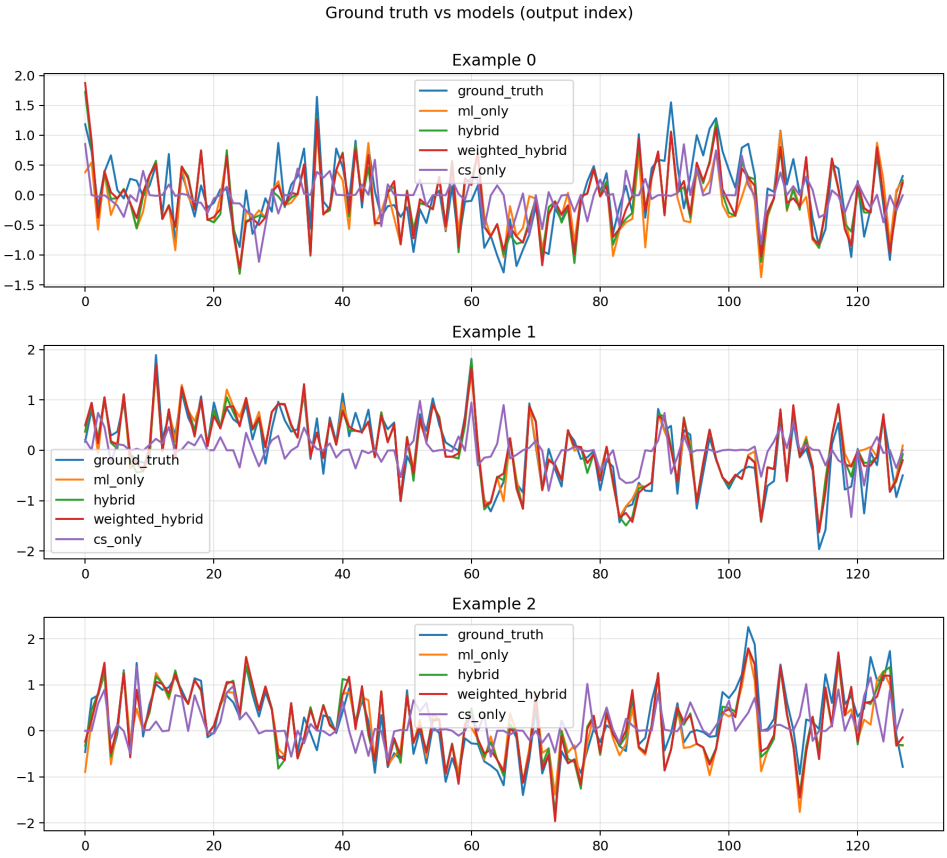


Figure 13: Example reconstructions along output index (Phase 0). ML-based methods track the ground truth; `cs_only` deviates toward an overly sparse profile.

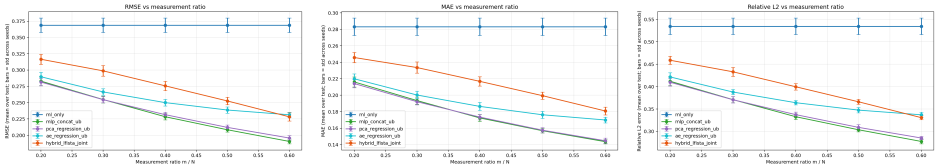


Figure 14: Direct $[u, b]$ benchmark (frozen run `20260422_102236`): test error versus ρ. Error bars: std across ten seeds.

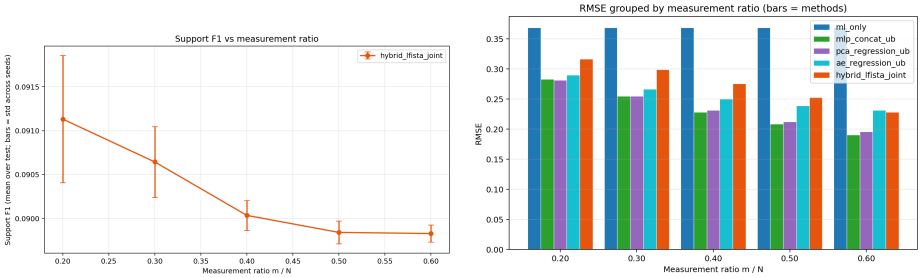


Figure 15: Left: support F1 for `hybrid_lfista_joint`. Right: grouped RMSE by ρ.

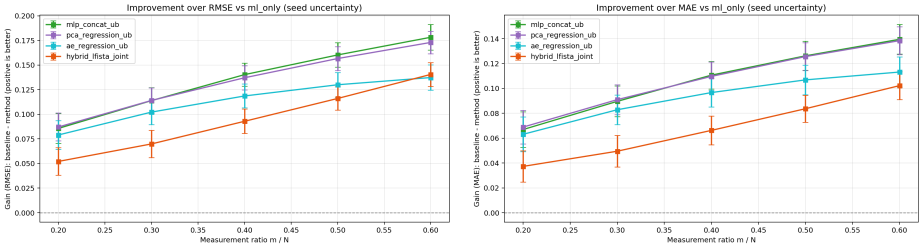


Figure 16: Gain over `ml_only` (positive is better). Direct methods that use b improve with ρ; `hybrid_lfista_joint` is included for comparison.

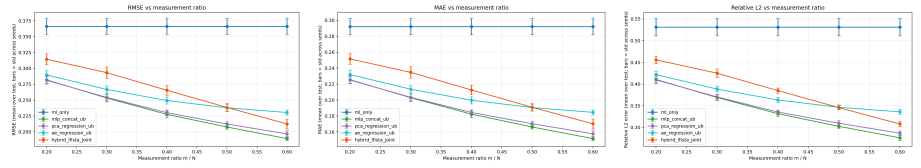


Figure 17: Direct $[u, b]$ joint-only benchmark with Ψ = DCT (frozen run `20260422_175210`): test error versus ρ. Same layout as Figure 14.

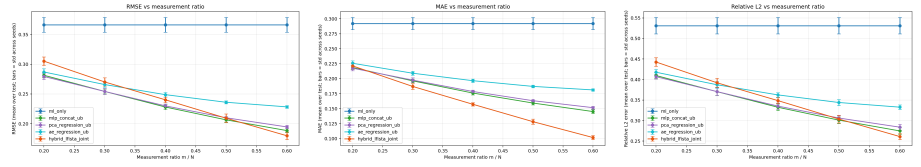


Figure 18: Run `20260422_205913` (Ψ = DCT, M subsample): test RMSE, MAE, and relative ℓ_2 vs. ρ (mean $\pm$ std over seeds). At medium–high ρ, `hybrid_lfista_joint` moves to the best or near-best error; the MAE panel shows the strongest separation from direct baselines.

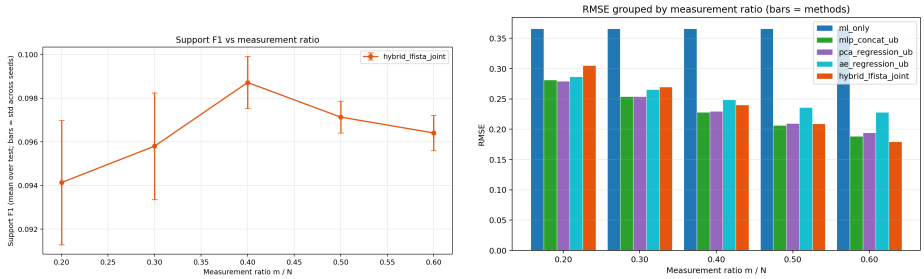


Figure 19: Left: support F1 for `hybrid_lfista_joint`. Right: grouped RMSE by ρ; at $\rho = 0.60$ the joint method has the lowest RMSE among the five methods (including `ml_only` in the script summary).

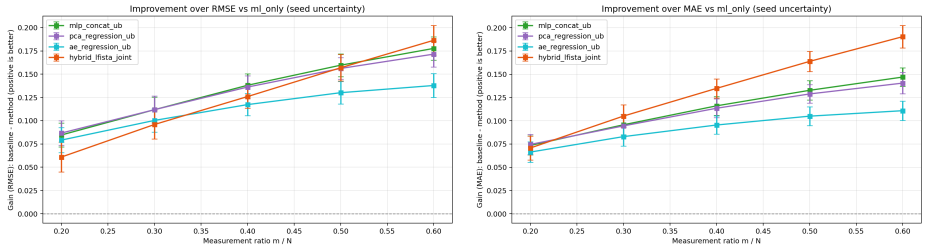


Figure 20: Gain over `ml_only` (positive is better) for run `20260422_205913`. The joint method shows the largest MAE gain across ρ; RMSE gain is strongest at the highest ρ.

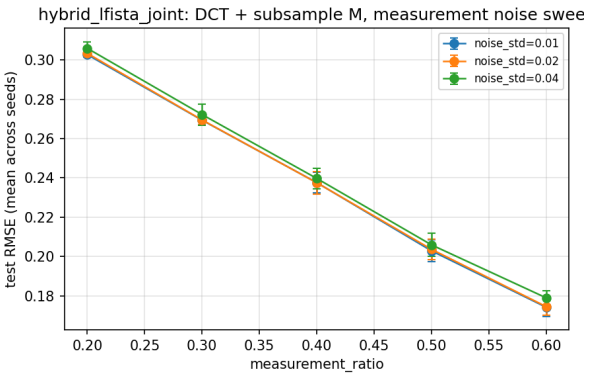


Figure 21: Pilot robustness: test RMSE (mean across seeds) for `hybrid_lfista_joint` vs. ρ at three values of measurement noise std on b; DCT Ψ, subsample M, three-seed profile (`--robustness-lite`). Curves shift upward with larger noise as expected; ordering across ρ is stable in this window.

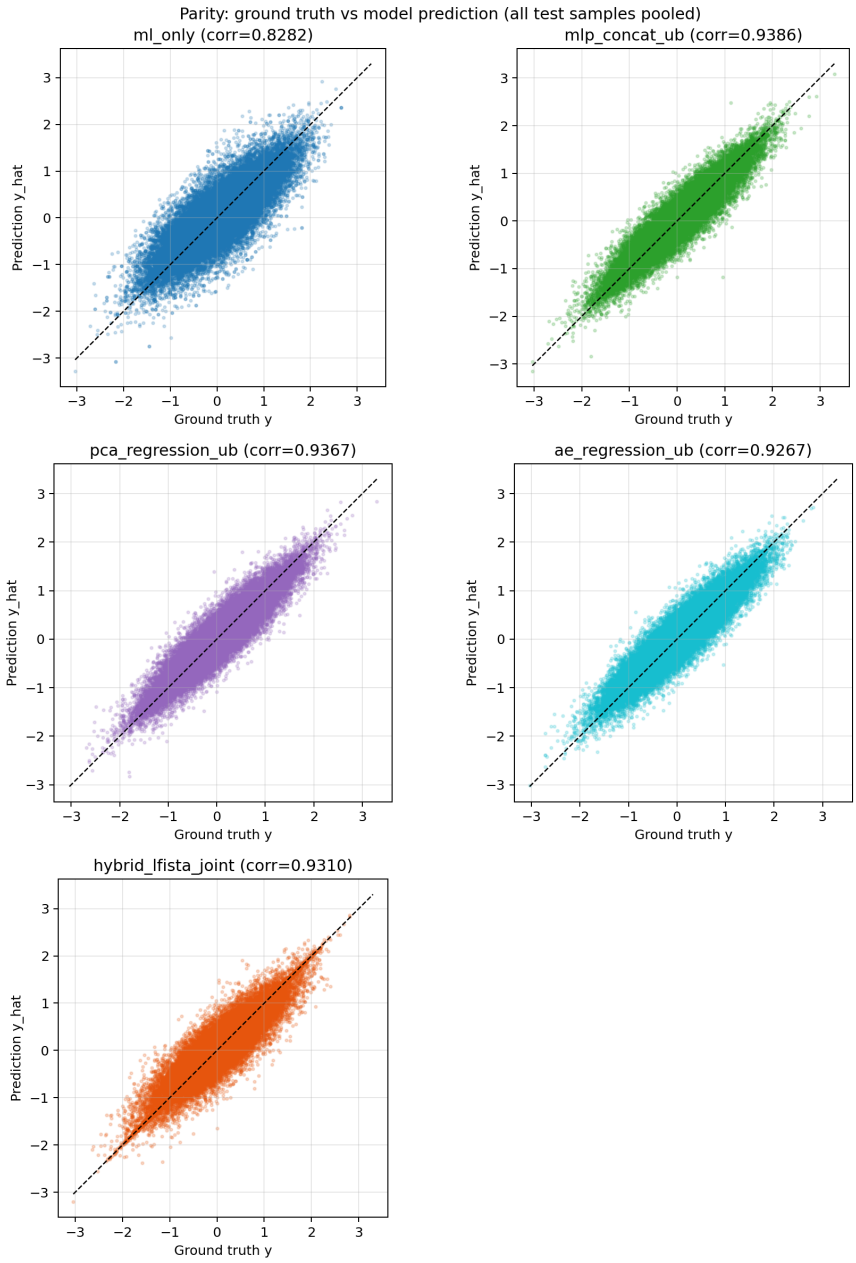


Figure 22: Pilot parity (pooled over all test output coordinates and all (seed, ρ) jobs in the arm): ground truth versus prediction for each method. Arm `noise_0p02_20260423_143832` with std(η) = 0.02 (same default scale as Config), DCT Ψ, subsample M, `--robustness-lite`. Correlation values are those printed by the benchmark script.

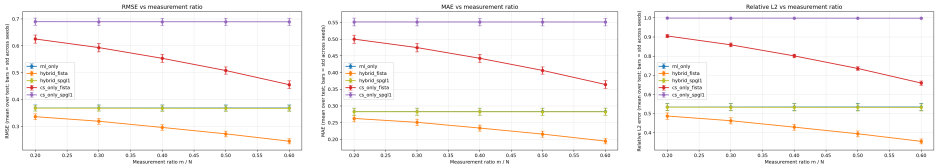


Figure 23: Step 1: RMSE, MAE, and relative ℓ_2 versus ρ for FISTA/SPGL1 variants. Error bars: std across seeds. `ml_only` is flat in ρ.

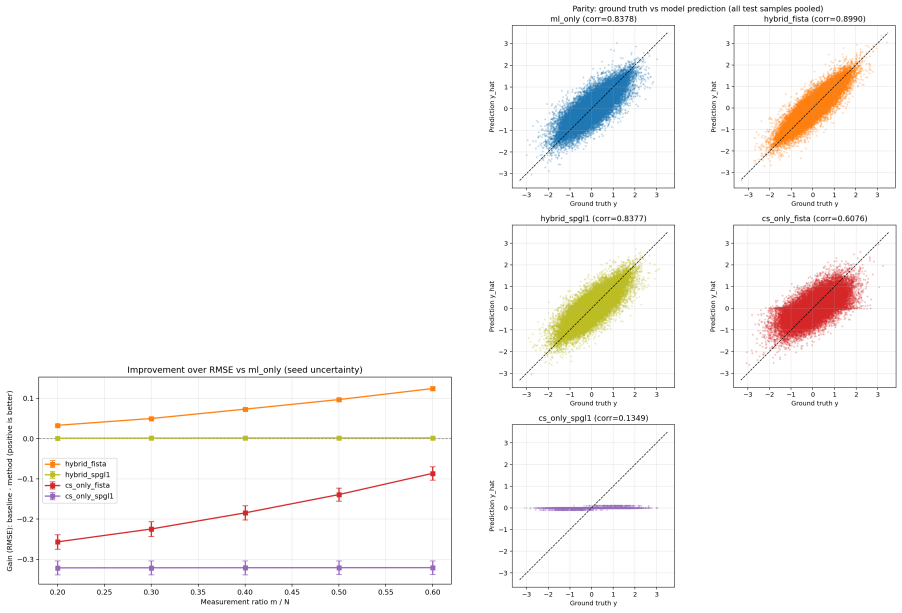


Figure 24: Step 1: RMSE gain over ml-only (left) and parity plot (right).

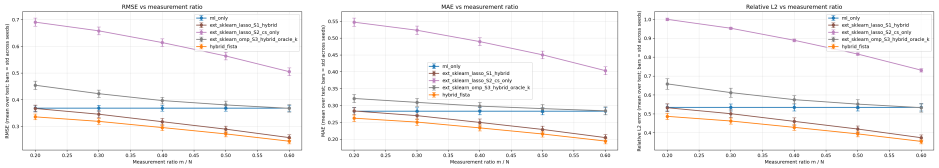


Figure 25: External baselines: RMSE, MAE, and relative ℓ_2 versus ρ. Error bars: std across seeds. `ml_only` is flat in ρ; `Lasso_S2` tracks near relative error one at low ρ.

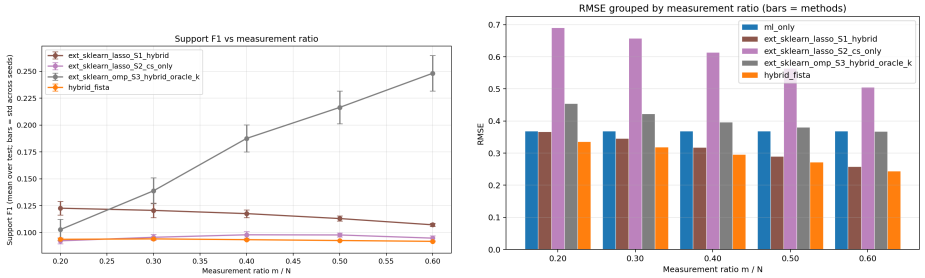


Figure 26: Support F1 (left) and grouped RMSE bars (right). OMP benefits from the oracle sparsity budget in F1; FISTA still leads in RMSE.

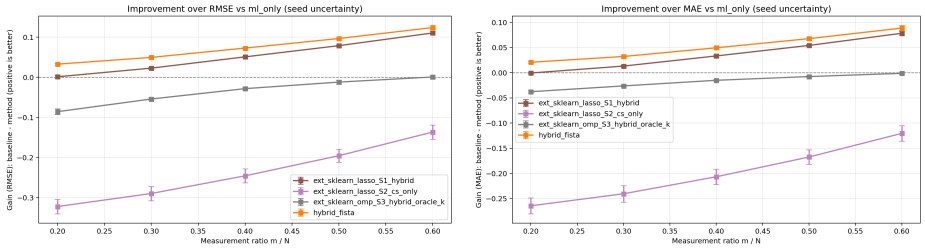
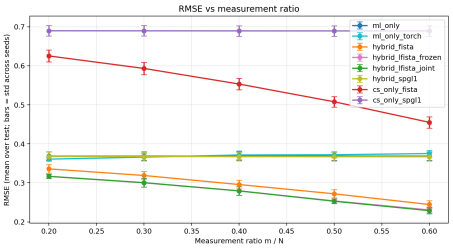
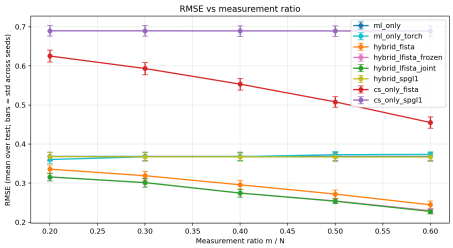


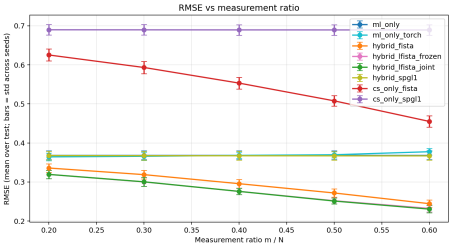
Figure 27: Gain over `ml_only` (positive is better): RMSE (left) and MAE (right).



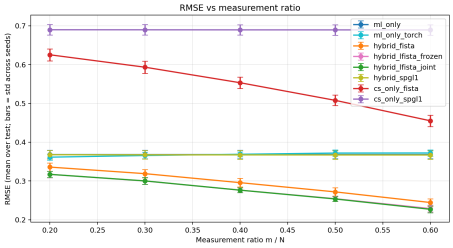
(a) Varying k at $k = 6$ (baseline match).



(b) Varying σ_η at $\sigma_\eta = 0.02$ (baseline).



(c) Varying innovation amplitude at 1.2 (baseline).



(d) Varying output noise std at 0.01 (baseline).

Figure 28: Representative RMSE versus $\rho = m/N$ curves under one-factor robustness sweeps. Methods include classical and LFISTA hybrids plus baselines from the same protocol.